

Презентация по лабораторной работе 8

Адресация IPv4 и IPv6. Настройка маршрутизации

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Содержание I

1. Информация
2. Вводная часть
3. Выполнение заданий
4. Выводы

Раздел 1

1. Информация

1.1 Докладчик

► Седохин Даниил Алексеевич

1.1 Докладчик

- ▶ Седохин Даниил Алексеевич
- ▶ Российский университет дружбы народов им. П. Лумумбы

Раздел 2

2. Вводная часть

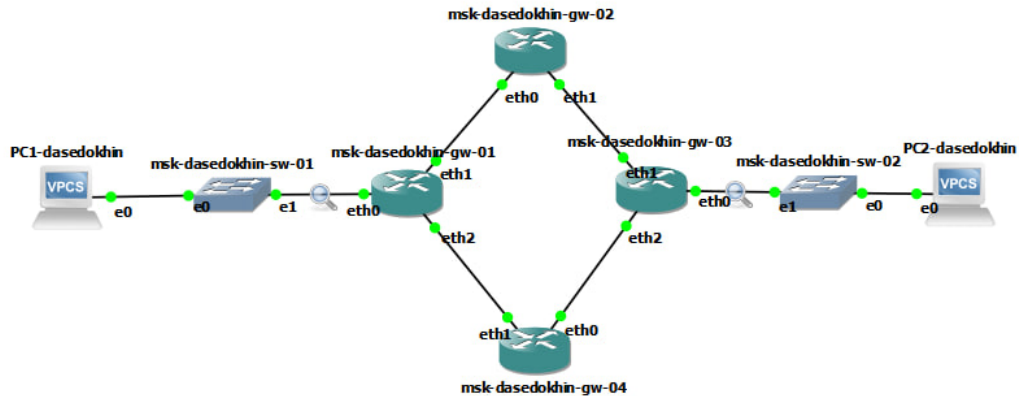
2.1 Цели и задачи

- ▶ Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.

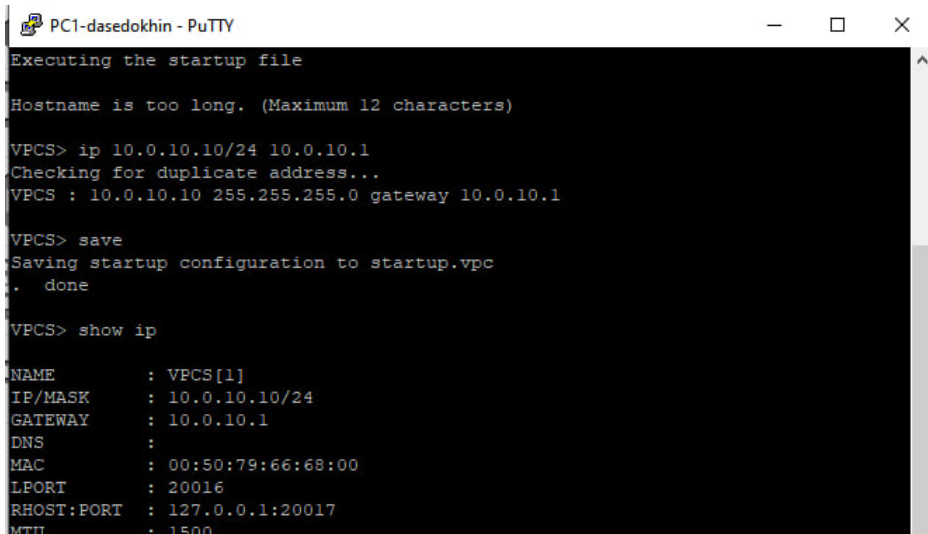
Раздел 3

3. Выполнение заданий

3.1 слайд 1



3.2 слайд 2



```
PC1-dasedokhin - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

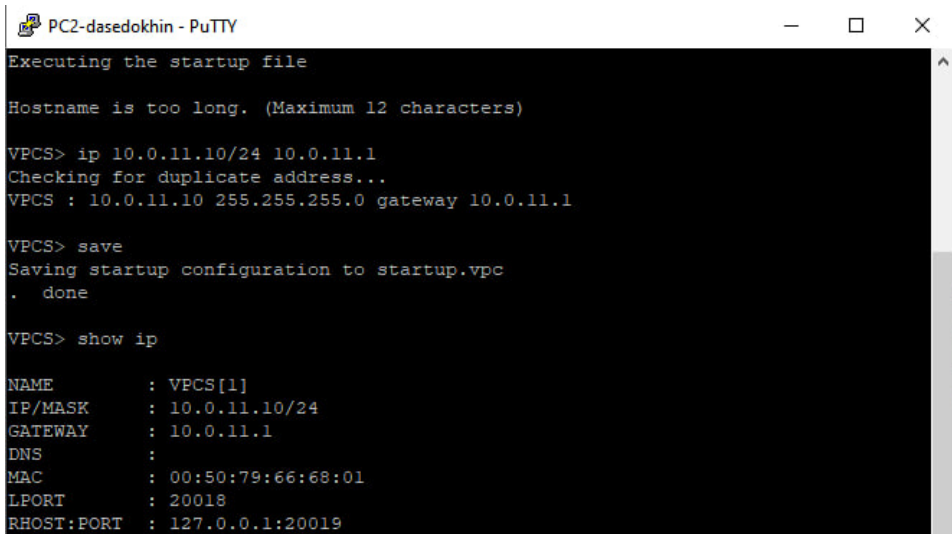
VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.10.10/24
GATEWAY    : 10.0.10.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20016
RHOST:PORT : 127.0.0.1:20017
MTU        : 1500
```

3.3 слайд 3



```
PC2-dasedokhin - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

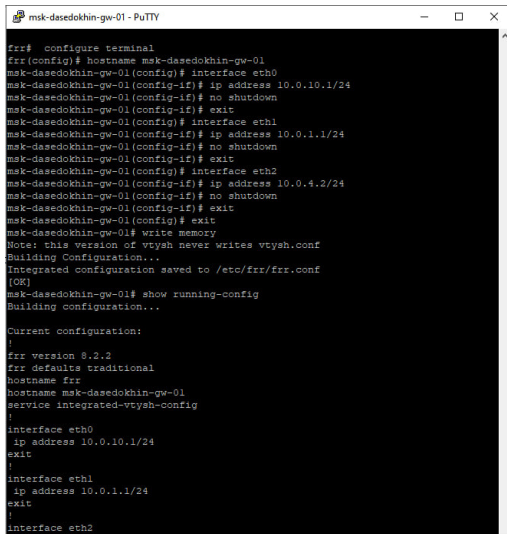
VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.11.10/24
GATEWAY    : 10.0.11.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 20018
RHOST:PORT : 127.0.0.1:20019
```

3.4 слайд 4

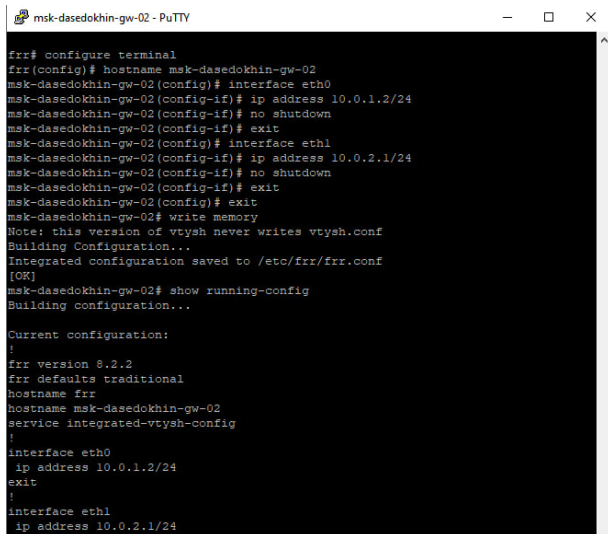


```
msk-dasedokhin-gw-01 - PuTTY

frr# configure terminal
frr(config)# hostname msk-dasedokhin-gw-01
msk-dasedokhin-gw-01(config)# interface eth0
msk-dasedokhin-gw-01(config-if)# ip address 10.0.10.1/24
msk-dasedokhin-gw-01(config-if)# no shutdown
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# interface eth1
msk-dasedokhin-gw-01(config-if)# ip address 10.0.1.1/24
msk-dasedokhin-gw-01(config-if)# no shutdown
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# interface eth2
msk-dasedokhin-gw-01(config-if)# ip address 10.0.4.2/24
msk-dasedokhin-gw-01(config-if)# no shutdown
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# exit
msk-dasedokhin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
{OK}
msk-dasedokhin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dasedokhin-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
```

3.5 слайд 5

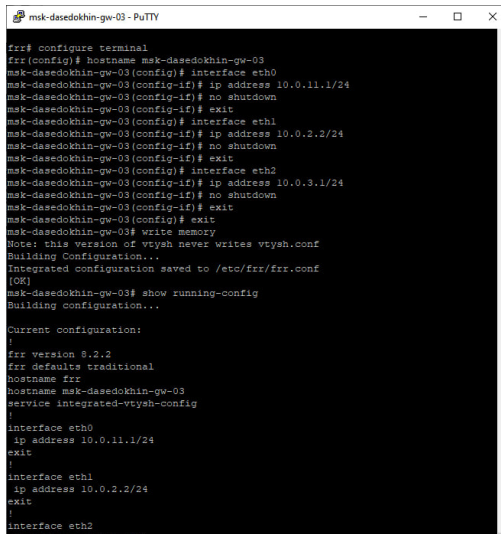


```
msk-dasedokhin-gw-02 - PuTTY

frr# configure terminal
frr(config)# hostname msk-dasedokhin-gw-02
msk-dasedokhin-gw-02(config)# interface eth0
msk-dasedokhin-gw-02(config-if)# ip address 10.0.1.2/24
msk-dasedokhin-gw-02(config-if)# no shutdown
msk-dasedokhin-gw-02(config-if)# exit
msk-dasedokhin-gw-02(config)# interface eth1
msk-dasedokhin-gw-02(config-if)# ip address 10.0.2.1/24
msk-dasedokhin-gw-02(config-if)# no shutdown
msk-dasedokhin-gw-02(config-if)# exit
msk-dasedokhin-gw-02(config)# exit
msk-dasedokhin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dasedokhin-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
exit
!
interface eth1
 ip address 10.0.2.1/24
```

3.6 слайд 6

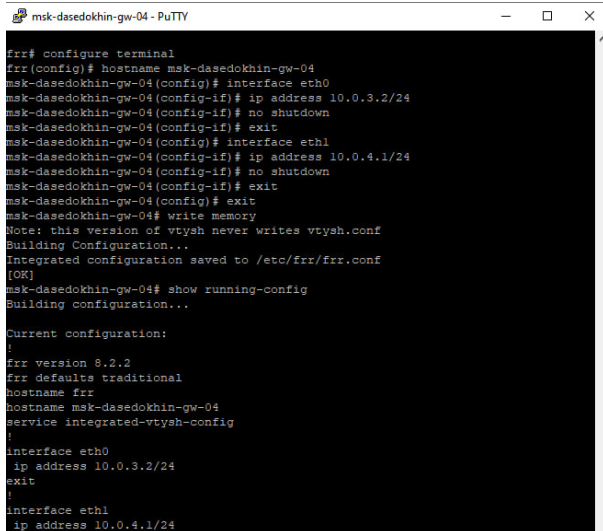


```
msk-dasedokhin-gw-03 - PuTTY

frr# configure terminal
frr(config)# hostname msk-dasedokhin-gw-03
msk-dasedokhin-gw-03(config)# interface eth0
msk-dasedokhin-gw-03(config-if)# ip address 10.0.11.1/24
msk-dasedokhin-gw-03(config-if)# no shutdown
msk-dasedokhin-gw-03(config-if)# exit
msk-dasedokhin-gw-03(config)# interface eth1
msk-dasedokhin-gw-03(config-if)# ip address 10.0.2.2/24
msk-dasedokhin-gw-03(config-if)# no shutdown
msk-dasedokhin-gw-03(config-if)# exit
msk-dasedokhin-gw-03(config)# interface eth2
msk-dasedokhin-gw-03(config-if)# ip address 10.0.3.1/24
msk-dasedokhin-gw-03(config-if)# no shutdown
msk-dasedokhin-gw-03(config-if)# exit
msk-dasedokhin-gw-03(config)# exit
msk-dasedokhin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dasedokhin-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
```

3.7 слайд 7

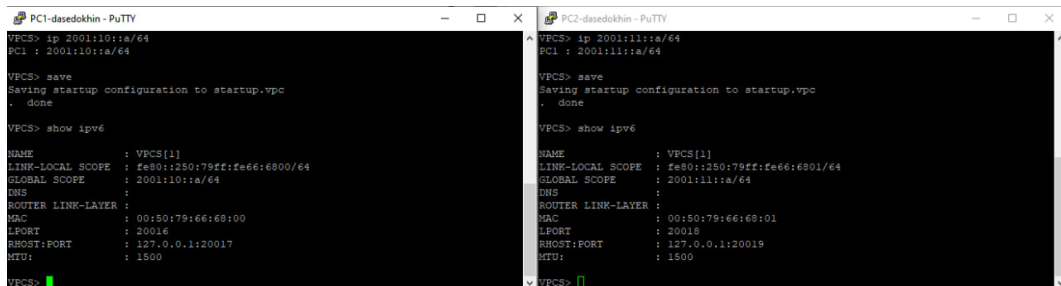


```
msk-dasedokhin-gw-04 - PuTTY

frr# configure terminal
frr(config)# hostname msk-dasedokhin-gw-04
msk-dasedokhin-gw-04(config)# interface eth0
msk-dasedokhin-gw-04(config-if)# ip address 10.0.3.2/24
msk-dasedokhin-gw-04(config-if)# no shutdown
msk-dasedokhin-gw-04(config-if)# exit
msk-dasedokhin-gw-04(config)# interface eth1
msk-dasedokhin-gw-04(config-if)# ip address 10.0.4.1/24
msk-dasedokhin-gw-04(config-if)# no shutdown
msk-dasedokhin-gw-04(config-if)# exit
msk-dasedokhin-gw-04(config)# exit
msk-dasedokhin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dasedokhin-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
```

3.8 слайд 8



The image shows two side-by-side PuTTY terminal windows. The left window is titled 'PC1-dasedokhin - PuTTY' and the right window is titled 'PC2-dasedokhin - PuTTY'. Both windows show the same sequence of commands and output for configuring IPv6 on a VPCS device.

```
VPCS> ip 2001:10::a/64
PC1 : 2001:10::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 2001:10::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 20016
RHOST:PORT          : 127.0.0.1:20017
MTU                 : 1500

VPCS>
```

The right window shows the same configuration but with the IP address 2001:11::a/64 instead of 2001:10::a/64.

```
VPCS> ip 2001:11::a/64
PC1 : 2001:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

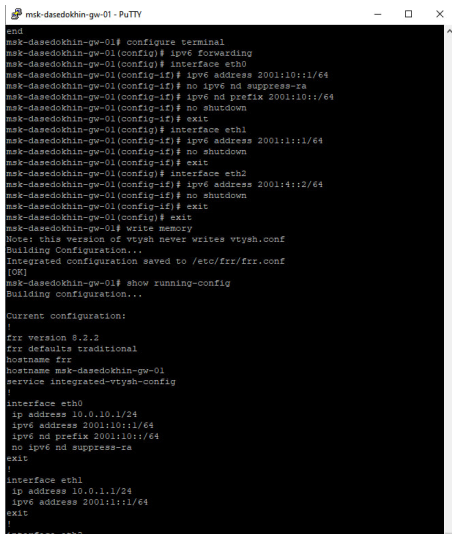
VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 20018
RHOST:PORT          : 127.0.0.1:20019
MTU                 : 1500

VPCS>
```

Рисунок 8: Присвоение IPv6 адресов для устройств PC1 и PC2

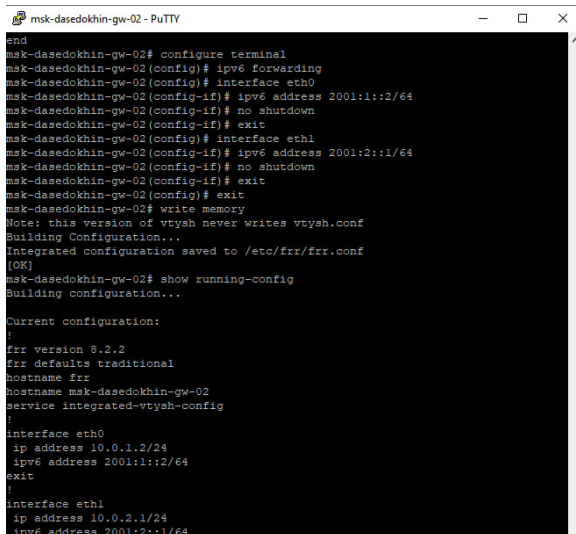
3.9 слайд 9



```
msk-dasedokhin-gw-01 - PuTTY
end
msk-dasedokhin-gw-01# configure terminal
msk-dasedokhin-gw-01(config)# ipv6 forwarding
msk-dasedokhin-gw-01(config)# interface eth0
msk-dasedokhin-gw-01(config-if)# ipv6 address 2001:10::1/64
msk-dasedokhin-gw-01(config-if)# no ipv6 nd suppress-ra
msk-dasedokhin-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msk-dasedokhin-gw-01(config-if)# no shutdown
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# interface eth1
msk-dasedokhin-gw-01(config-if)# ipv6 address 2001:1::1/64
msk-dasedokhin-gw-01(config-if)# no shutdown
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# interface eth2
msk-dasedokhin-gw-01(config-if)# ipv6 address 2001:4::2/64
msk-dasedokhin-gw-01(config-if)# no shutdown
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# exit
msk-dasedokhin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dasedokhin-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
 ipv6 address 2001:10::1/64
 ipv6 nd prefix 2001:10::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.1.1/24
 ipv6 address 2001:1::1/64
exit
!
interface eth2
```

3.10 слайд 10



```
msk-dasedokhin-gw-02# end
msk-dasedokhin-gw-02# configure terminal
msk-dasedokhin-gw-02(config)# ipv6 forwarding
msk-dasedokhin-gw-02(config)# interface eth0
msk-dasedokhin-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-dasedokhin-gw-02(config-if)# no shutdown
msk-dasedokhin-gw-02(config-if)# exit
msk-dasedokhin-gw-02(config)# interface eth1
msk-dasedokhin-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-dasedokhin-gw-02(config-if)# no shutdown
msk-dasedokhin-gw-02(config-if)# exit
msk-dasedokhin-gw-02(config)# exit
msk-dasedokhin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-02# show running-config
Building configuration...

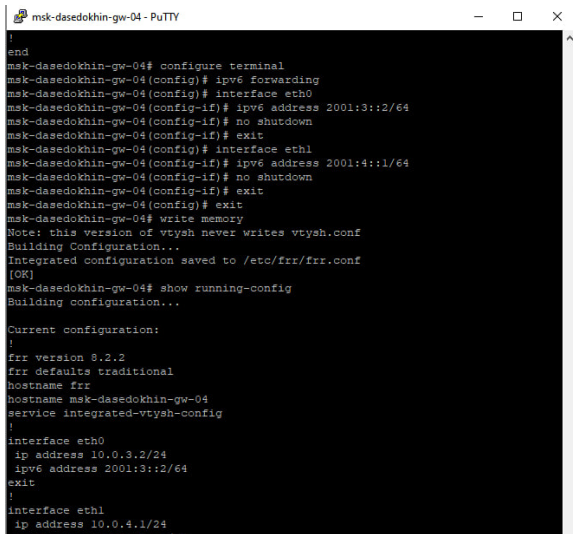
Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dasedokhin-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ip address 10.0.2.1/24
 ipv6 address 2001:2::1/64
```

3.11 слайд 11

```
mks-dasedokhin-gw-03 - PuTTY
end
mks-dasedokhin-gw-03# configure terminal
mks-dasedokhin-gw-03(config)# ipv6 forwarding
mks-dasedokhin-gw-03(config)# interface eth0
mks-dasedokhin-gw-03(config-if)# ipv6 address 2001:11::1/64
mks-dasedokhin-gw-03(config-if)# no ipv6 nd suppress-ra
mks-dasedokhin-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
mks-dasedokhin-gw-03(config-if)# no shutdown
mks-dasedokhin-gw-03(config-if)# exit
mks-dasedokhin-gw-03(config)# interface eth1
mks-dasedokhin-gw-03(config-if)# ipv6 address 2001:2::2/64
mks-dasedokhin-gw-03(config-if)# no shutdown
mks-dasedokhin-gw-03(config-if)# exit
mks-dasedokhin-gw-03(config)# interface eth2
mks-dasedokhin-gw-03(config-if)# ipv6 address 2001:3::1/64
mks-dasedokhin-gw-03(config-if)# no shutdown
mks-dasedokhin-gw-03(config-if)# exit
mks-dasedokhin-gw-03(config)# exit
mks-dasedokhin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
mks-dasedokhin-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname mks-dasedokhin-gw-03
service integrated-vtysh-conf
!
interface eth0
 ip address 10.0.11.1/24
 ipv6 address 2001:11::1/64
 ipv6 nd prefix 2001:11::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.2.2/24
 ipv6 address 2001:2::2/64
exit
!
```

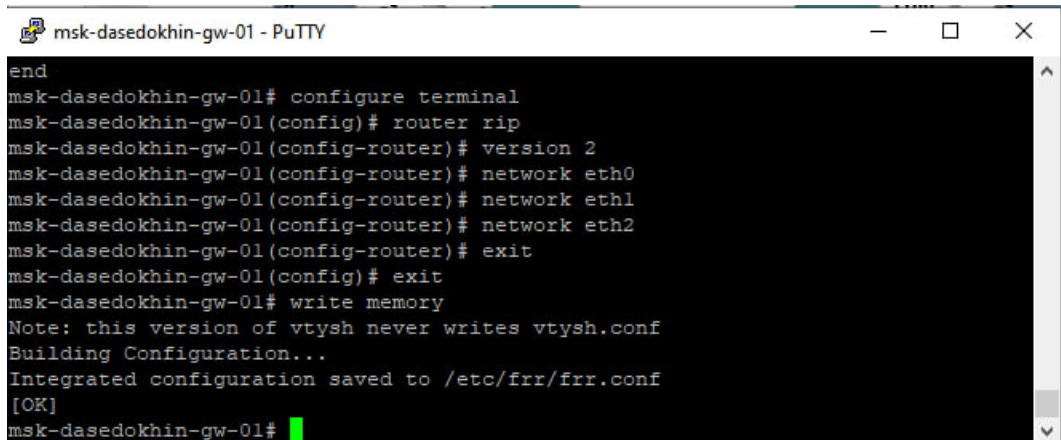
3.12 слайд 12



```
msk-dasedokhin-gw-04 - PuTTY
!
end
msk-dasedokhin-gw-04# configure terminal
msk-dasedokhin-gw-04(config)# ipv6 forwarding
msk-dasedokhin-gw-04(config)# interface eth0
msk-dasedokhin-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-dasedokhin-gw-04(config-if)# no shutdown
msk-dasedokhin-gw-04(config-if)# exit
msk-dasedokhin-gw-04(config)# interface eth1
msk-dasedokhin-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-dasedokhin-gw-04(config-if)# no shutdown
msk-dasedokhin-gw-04(config-if)# exit
msk-dasedokhin-gw-04(config)# exit
msk-dasedokhin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dasedokhin-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
exit
!
interface eth1
 ip address 10.0.4.1/24
```

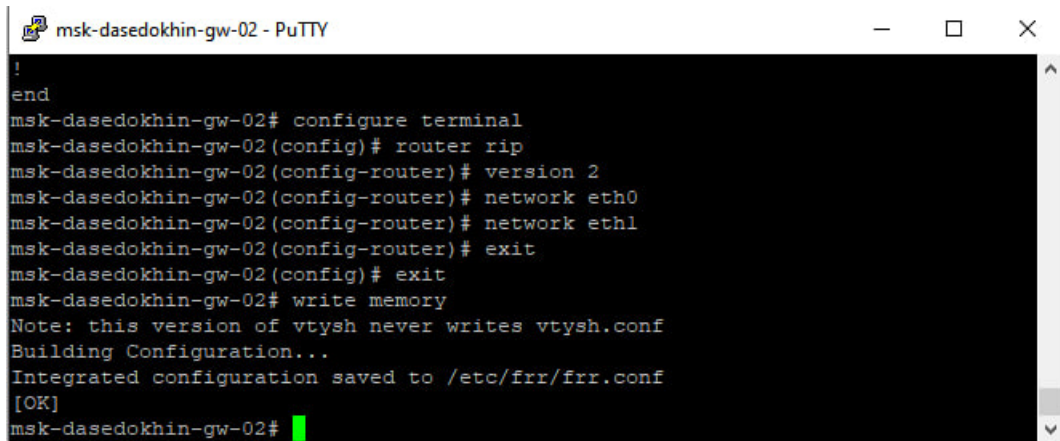
3.13 слайд 13



```
msk-dasedokhin-gw-01# end
msk-dasedokhin-gw-01# configure terminal
msk-dasedokhin-gw-01(config)# router rip
msk-dasedokhin-gw-01(config-router)# version 2
msk-dasedokhin-gw-01(config-router)# network eth0
msk-dasedokhin-gw-01(config-router)# network eth1
msk-dasedokhin-gw-01(config-router)# network eth2
msk-dasedokhin-gw-01(config-router)# exit
msk-dasedokhin-gw-01(config)# exit
msk-dasedokhin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-01#
```

Рисунок 13: Настройка RIP на маршрутизаторе gw-01

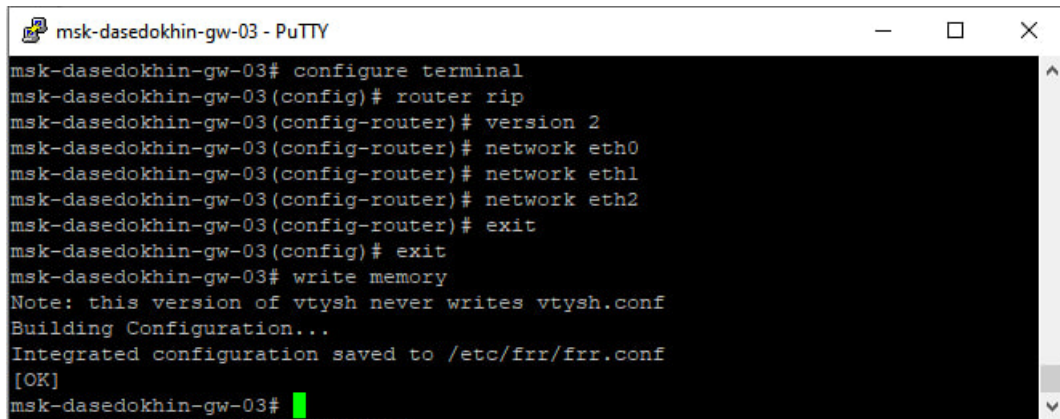
3.14 слайд 14



```
msk-dasedokhin-gw-02 - PuTTY
!
end
msk-dasedokhin-gw-02# configure terminal
msk-dasedokhin-gw-02 (config)# router rip
msk-dasedokhin-gw-02 (config-router)# version 2
msk-dasedokhin-gw-02 (config-router)# network eth0
msk-dasedokhin-gw-02 (config-router)# network eth1
msk-dasedokhin-gw-02 (config-router)# exit
msk-dasedokhin-gw-02 (config)# exit
msk-dasedokhin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-02#
```

Рисунок 14: Настройка RIP на маршрутизаторе gw-02

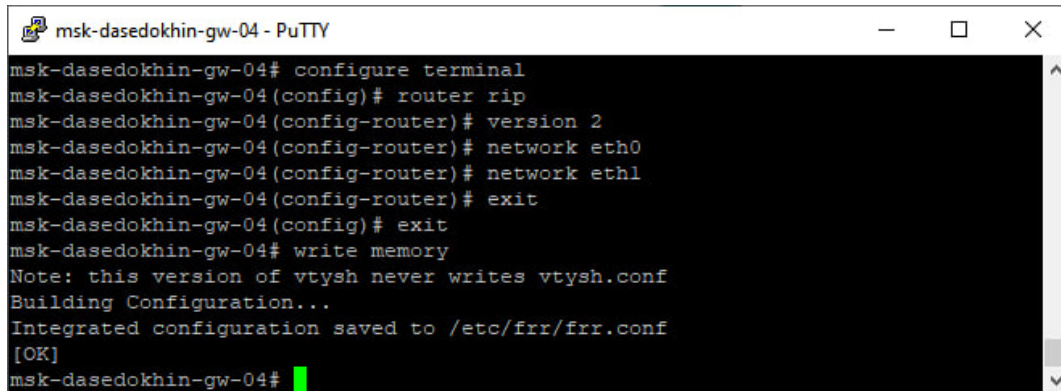
3.15 слайд 15



```
msk-dasedokhin-gw-03 - PuTTY
msk-dasedokhin-gw-03# configure terminal
msk-dasedokhin-gw-03 (config)# router rip
msk-dasedokhin-gw-03 (config-router)# version 2
msk-dasedokhin-gw-03 (config-router)# network eth0
msk-dasedokhin-gw-03 (config-router)# network eth1
msk-dasedokhin-gw-03 (config-router)# network eth2
msk-dasedokhin-gw-03 (config-router)# exit
msk-dasedokhin-gw-03 (config)# exit
msk-dasedokhin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-03#
```

Рисунок 15: Настройка RIP на маршрутизаторе gw-03

3.16 слайд 16



```
msk-dasedokhin-gw-04# configure terminal
msk-dasedokhin-gw-04(config)# router rip
msk-dasedokhin-gw-04(config-router)# version 2
msk-dasedokhin-gw-04(config-router)# network eth0
msk-dasedokhin-gw-04(config-router)# network eth1
msk-dasedokhin-gw-04(config-router)# exit
msk-dasedokhin-gw-04(config)# exit
msk-dasedokhin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-04#
```

Рисунок 16: Настройка RIP на маршрутизаторе gw-04

3.17 слайд 17

```

msk-dasedokhin-gw-01 - PuTTY
msk-dasedokhin-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
        O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
        T - Table, v - VNC, V - VNC-Direct, A - Babel, F - FBR,
        f - OpenFabric,
        > - selected route, * - FIB route, q - queued, r - rejected, b - backup
        t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/2] via 10.0.1.2, eth1, weight 1, 00:02:02
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:00:21
R>* 10.0.11.0/24 [120/3] via 10.0.1.2, eth1, weight 1, 00:01:06
msk-dasedokhin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network      Next Hop      Metric From      Tag Time
C(i) 10.0.1.0/24    0.0.0.0        1 self           0
R(n) 10.0.2.0/24    10.0.1.2       2 10.0.1.2        0 02:45
R(n) 10.0.3.0/24    10.0.4.1       2 10.0.4.1        0 02:33
C(i) 10.0.4.0/24    0.0.0.0        1 self           0
C(i) 10.0.10.0/24   0.0.0.0        1 self           0
R(n) 10.0.11.0/24   10.0.1.2       3 10.0.1.2        0 02:45
msk-dasedokhin-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 25 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send Recv  Key-chain
    eth0            2     2
    eth1            2     2
    eth2            2     2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:

```

3.18 слайд 18

```

msk-dasedokhin-gw-02 - PuTTY
msk-dasedokhin-gw-02# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.3.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:02:10
R>* 10.0.4.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:02:53
R>* 10.0.10.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:02:53
R>* 10.0.11.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:01:55
msk-dasedokhin-gw-02# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24        0.0.0.0            1 self                0
C(i) 10.0.2.0/24        0.0.0.0            1 self                0
R(n) 10.0.3.0/24        10.0.2.2            2 10.0.2.2            0 02:41
R(n) 10.0.4.0/24        10.0.1.1            2 10.0.1.1            0 02:43
R(n) 10.0.10.0/24       10.0.1.1            2 10.0.1.1            0 02:43
R(n) 10.0.11.0/24       10.0.2.2            2 10.0.2.2            0 02:41
msk-dasedokhin-gw-02# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 11 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface          Send Recv   Key-chain
    eth0                2     2
    eth1                2     2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:

```

3.19 слайд 19

```

msk-dasedokhin-gw-03 - PuTTY
msk-dasedokhin-gw-03# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - FBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.2.1, eth1, weight 1, 00:02:47
R>* 10.0.4.0/24 [120/2] via 10.0.3.2, eth2, weight 1, 00:01:56
R>* 10.0.10.0/24 [120/3] via 10.0.2.1, eth1, weight 1, 00:02:47
msk-dasedokhin-gw-03# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network          Next Hop        Metric From        Tag Time
R(n) 10.0.1.0/24       10.0.2.1          2 10.0.2.1          0 02:43
C(i) 10.0.2.0/24       0.0.0.0           1 self              0
C(i) 10.0.3.0/24       0.0.0.0           1 self              0
R(n) 10.0.4.0/24       10.0.3.2          2 10.0.3.2          0 02:41
R(n) 10.0.10.0/24      10.0.2.1          3 10.0.2.1          0 02:43
C(i) 10.0.11.0/24      0.0.0.0           1 self              0
msk-dasedokhin-gw-03# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 21 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface    Send  Recv  Key-chain
    eth0         2     2
    eth1         2     2
    eth2         2     2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:

```

3.20 слайд 20

```

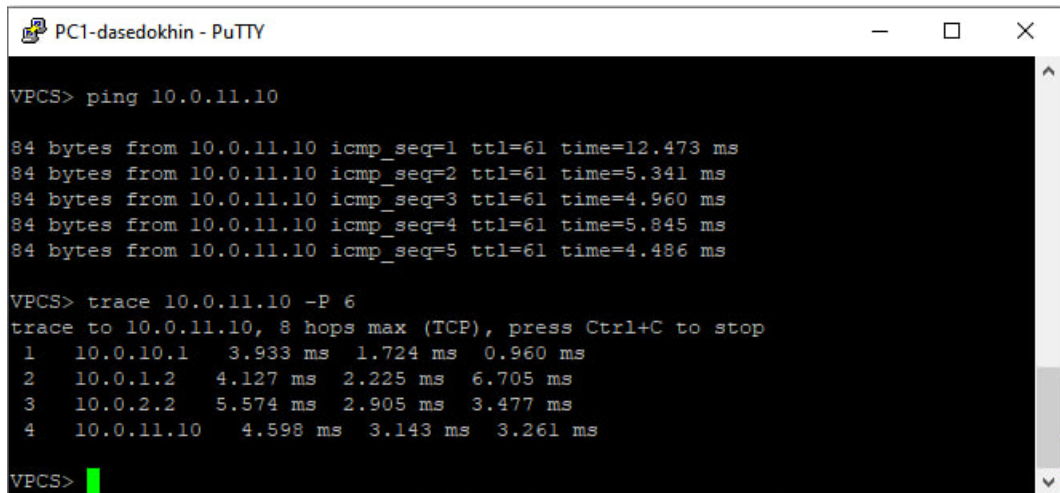
msk-dasedokhin-gw-04 - PuTTY
msk-dasedokhin-gw-04# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:02:34
R>* 10.0.2.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:02:38
R>* 10.0.10.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:02:34
R>* 10.0.11.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:02:38
msk-dasedokhin-gw-04# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop           Metric From       Tag Time
R(n) 10.0.1.0/24       10.0.4.2           2 10.0.4.2        0 02:36
R(n) 10.0.2.0/24       10.0.3.1           2 10.0.3.1        0 02:53
C(i) 10.0.3.0/24       0.0.0.0            1 self            0
C(i) 10.0.4.0/24       0.0.0.0            1 self            0
R(n) 10.0.10.0/24      10.0.4.2           2 10.0.4.2        0 02:36
R(n) 10.0.11.0/24      10.0.3.1           2 10.0.3.1        0 02:53
msk-dasedokhin-gw-04# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 2 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send Recv  Key-chain
    eth0            2      2
    eth1            2      2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:

```

3.21 слайд 21



```
PC1-dasedokhin - PuTTY

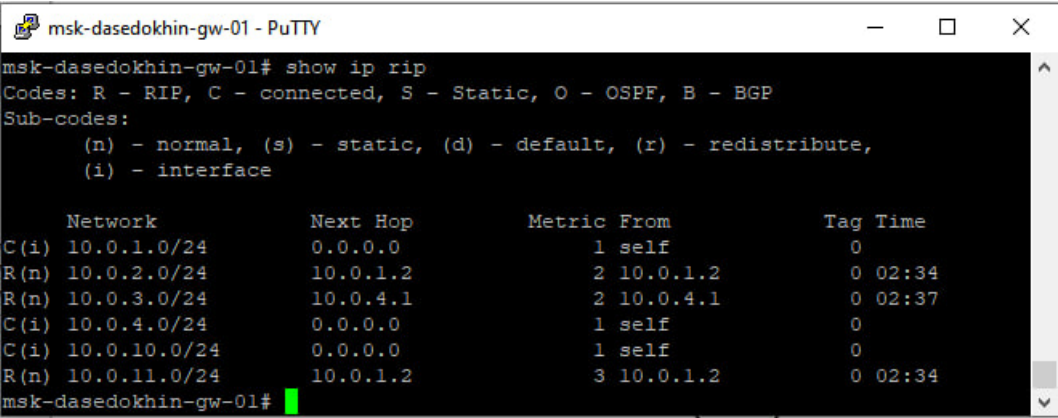
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=12.473 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=5.341 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=4.960 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=5.845 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=4.486 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1   3.933 ms  1.724 ms  0.960 ms
 2  10.0.1.2    4.127 ms  2.225 ms  6.705 ms
 3  10.0.2.2    5.574 ms  2.905 ms  3.477 ms
 4  10.0.11.10  4.598 ms  3.143 ms  3.261 ms

VPCS> 
```

3.22 слайд 22



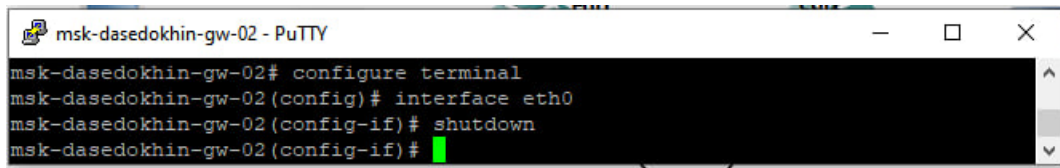
msk-dasedokhin-gw-01 - PuTTY

```
msk-dasedokhin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop           Metric From       Tag Time
C(i) 10.0.1.0/24       0.0.0.0            1 self            0
R(n) 10.0.2.0/24       10.0.1.2           2 10.0.1.2         0 02:34
R(n) 10.0.3.0/24       10.0.4.1           2 10.0.4.1         0 02:37
C(i) 10.0.4.0/24       0.0.0.0            1 self            0
C(i) 10.0.10.0/24      0.0.0.0            1 self            0
R(n) 10.0.11.0/24      10.0.1.2           3 10.0.1.2         0 02:34
msk-dasedokhin-gw-01#
```

Рисунок 22: Проверка метрики протокола RIP

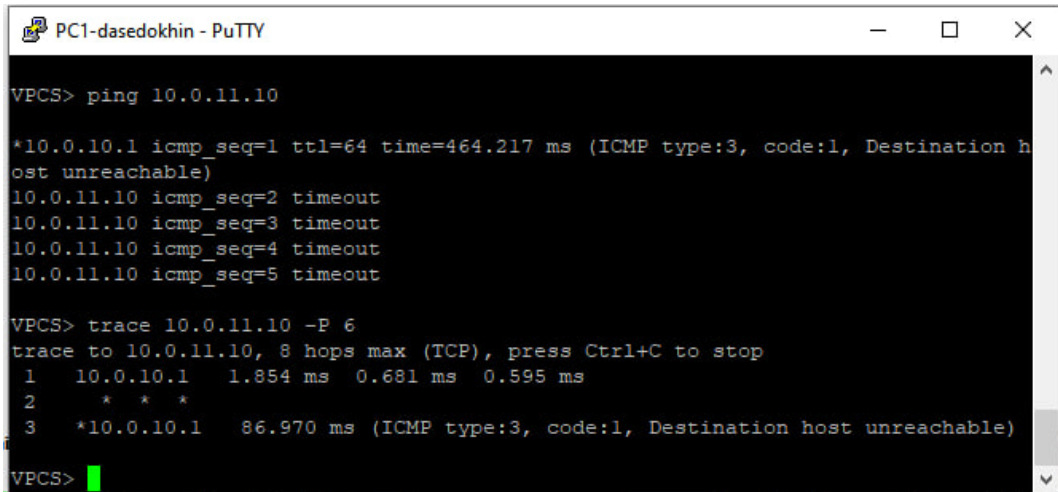
3.23 слайд 23



```
msk-dasedokhin-gw-02 - PuTTY
msk-dasedokhin-gw-02# configure terminal
msk-dasedokhin-gw-02 (config)# interface eth0
msk-dasedokhin-gw-02 (config-if)# shutdown
msk-dasedokhin-gw-02 (config-if)#
```

Рисунок 23: Отключение интерфейса на маршрутизаторе gw-02

3.24 слайд 24



```
PC1-dasedokhin - PuTTY

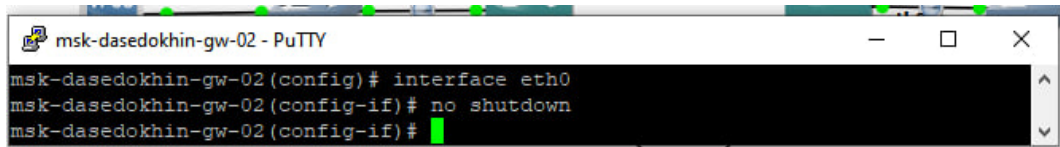
VPCS> ping 10.0.11.10

*10.0.10.1 icmp_seq=1 ttl=64 time=464.217 ms (ICMP type:3, code:1, Destination host unreachable)
10.0.11.10 icmp_seq=2 timeout
10.0.11.10 icmp_seq=3 timeout
10.0.11.10 icmp_seq=4 timeout
10.0.11.10 icmp_seq=5 timeout

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1   1.854 ms  0.681 ms  0.595 ms
 2      *    *    *
 3  *10.0.10.1  86.970 ms (ICMP type:3, code:1, Destination host unreachable)

VPCS>
```

3.25 слайд 25



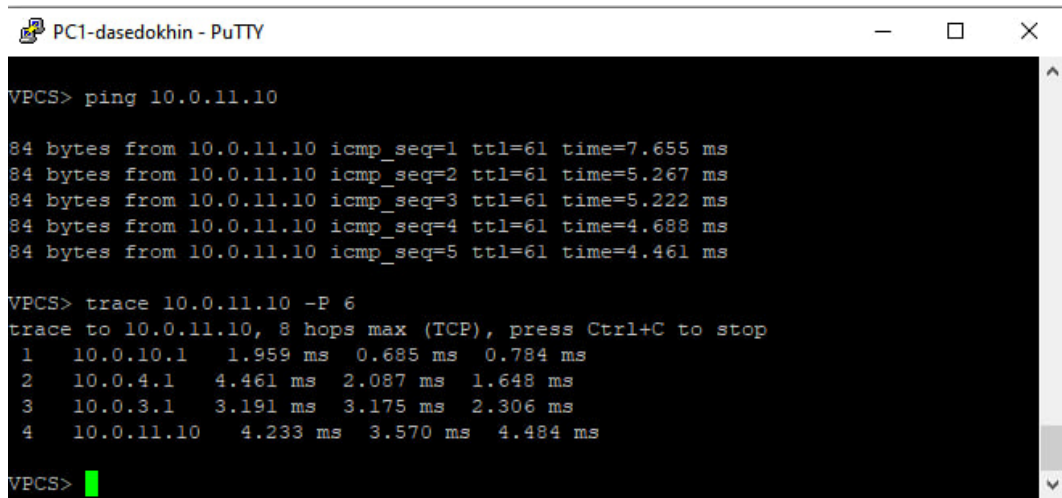
The image shows a PuTTY terminal window titled "msk-dasedokhin-gw-02 - PuTTY". The terminal output shows the following commands and prompts:

```
msk-dasedokhin-gw-02 (config)# interface eth0
msk-dasedokhin-gw-02 (config-if)# no shutdown
msk-dasedokhin-gw-02 (config-if)#
```

The cursor is positioned at the end of the third line, ready for the next command.

Рисунок 25: Включение интерфейса на маршрутизаторе gw-02

3.26 слайд 26



```
PC1-dasedokhin - PuTTY

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=7.655 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=5.267 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=5.222 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=4.688 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=4.461 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    1.959 ms  0.685 ms  0.784 ms
 2  10.0.4.1     4.461 ms  2.087 ms  1.648 ms
 3  10.0.3.1     3.191 ms  3.175 ms  2.306 ms
 4  10.0.11.10   4.233 ms  3.570 ms  4.484 ms

VPCS> 
```

3.27 слайд 27

Захват из Standard input [mk-dashed00in-gw-01 (ethnet)] to mk-dashed00in-gw-01 (ethnet)

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения: < Cid-/>

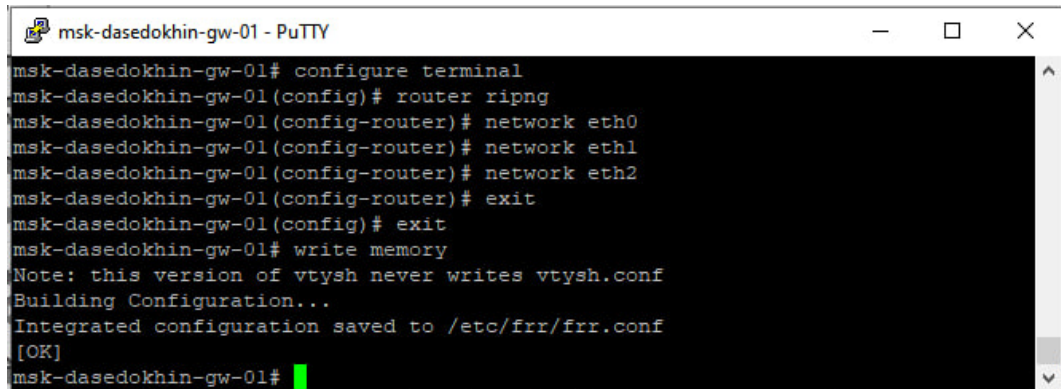
No.	Time	Source	Destination	Protocol	Length	Info
240	1773.593897	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
241	1773.594886	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 23953 → 23954 [SVN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766073314 TSecr=0 WS=2
242	1773.596288	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
243	1773.597867	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 23953 → 23954 [SVN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766073314 TSecr=0 WS=2
244	1773.600914	10.0.3.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
245	1773.601434	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 23953 → 23954 [SVN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766073314 TSecr=0 WS=2
246	1773.604311	10.0.3.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
247	1773.605674	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 23953 → 23954 [SVN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766073314 TSecr=0 WS=2
248	1773.607797	10.0.3.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
249	1773.608150	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 23953 → 23954 [SVN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766073314 TSecr=0 WS=2
250	1773.612225	10.0.10.10	10.0.10.10	TCP	0	[23954 → 23953] [SVN, ACK] Seq=0 Win=0 Len=0
251	1773.614506	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 23953 → 23954 [SVN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766073314 TSecr=0 WS=2
252	1773.617835	10.0.10.10	10.0.10.10	TCP	60	[TCP Port numbers reused] 23954 → 23953 [SVN, ACK] Seq=0 Ack=1 Win=0 Len=0
253	1773.620030	10.0.10.10	10.0.11.10	TCP	74	23953 → 23954 [SVN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766073314 TSecr=0 WS=2
254	1773.624107	10.0.11.10	10.0.10.10	TCP	60	[TCP Port numbers reused] 23954 → 23953 [SVN, ACK] Seq=0 Ack=1 Win=0 Len=0
255	1000.743273	10.0.10.1	224.0.0.0	IGMPv2	146	Response

* Frame 248: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0

- Section number: 1
- Interface id: 0 (-)
- Encapsulation type: Ethernet (1)
- Arrival Time: Dec 18, 2025 18:55:14.200212000 RTZ 2 (sma)
- UTC Arrival Time: Dec 18, 2025 15:55:14.200212000 UTC
- Epoch Arrival Time: 1766073314.200212000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 0.002123000 seconds]
- [Time delta from previous displayed frame: 0.002123000 seconds]
- [Time since reference or first frame: 1773.607797000 seconds]
- Frame Number: 248
- Frame Length: 102 bytes (816 bits)
- Capture Length: 102 bytes (816 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: ethertypes:ip:icmp:ip:tcp]
- [Coloring Rule Name: ICMP errors]
- [Coloring Rule String: icmp.type in { 3, 5, 11 } || icmp.v.type in { 1, 4 }]
- Ethernet II, Src: 0c:14:a2:19:00:00 (0c:14:a2:19:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
 - Destination: Private_66:68:00 (00:50:79:66:68:00)
 - Source: 0c:14:a2:19:00:00 (0c:14:a2:19:00:00)
 - Type: IPv4 (0x0800)
 - [Stream index: 8]
- Internet Protocol Version 4, Src: 10.0.3.1, Dst: 10.0.10.10
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
 - Total length: 80
 - Identification: 0x357d (13693)
 - 000. = Flags: 0x0
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 62
 - Protocol: ICMP (1)
 - Header checksum: 0x255e (validation disabled)
 - [Header checksum status: Unverified]
 - Source Address: 10.0.3.1
 - Destination Address: 10.0.10.10
 - [Stream index: 7]

0000 00 50 79 66 68 00 0c 14 a2 19 00 00
0010 00 50 35 7d 00 00 3e 01 25 5e 0a 00
0020 0a 0a 00 00 1e 42 00 00 00 00 45 00
0030 40 00 01 06 50 a1 0a 00 0a 0a 0a 00
0040 5d 92 69 44 23 e2 00 00 00 00 00 00
0050 00 00 02 04 05 04 01 01 00 0a 69 00
0060 00 00 01 03 03 01

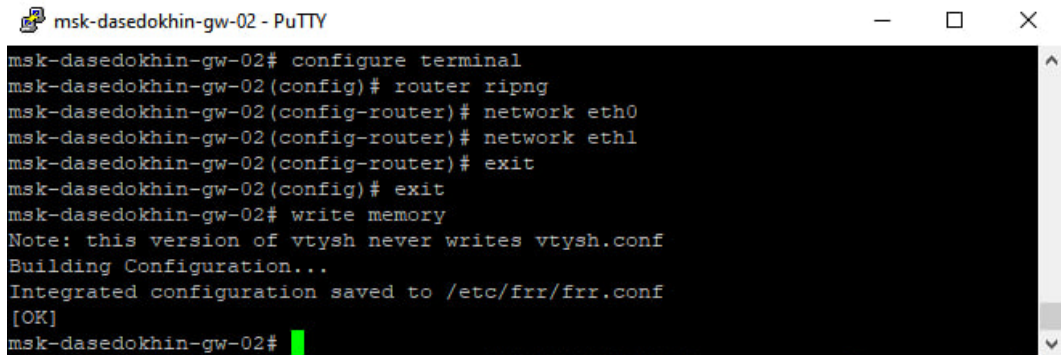
3.28 слайд 28



```
msk-dasedokhin-gw-01# configure terminal
msk-dasedokhin-gw-01(config)# router ripng
msk-dasedokhin-gw-01(config-router)# network eth0
msk-dasedokhin-gw-01(config-router)# network eth1
msk-dasedokhin-gw-01(config-router)# network eth2
msk-dasedokhin-gw-01(config-router)# exit
msk-dasedokhin-gw-01(config)# exit
msk-dasedokhin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-01#
```

Рисунок 28: Настройка RIPng для сетей IPv6 на маршрутизаторе gw-01

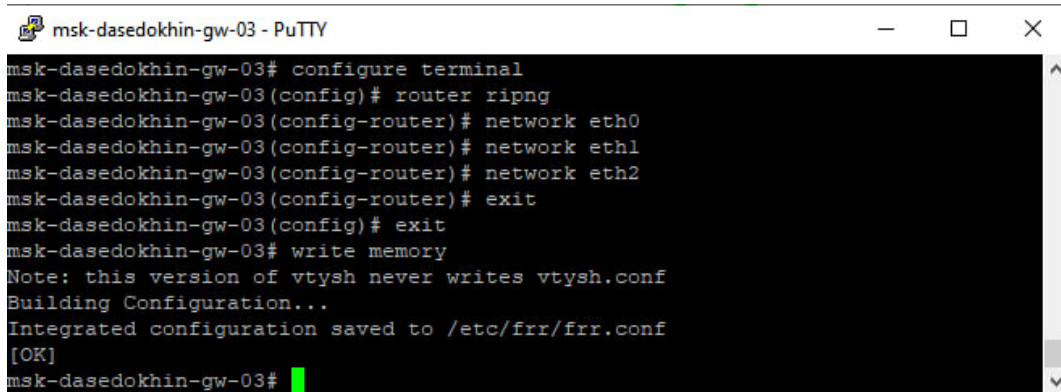
3.29 слайд 29



```
msk-dasedokhin-gw-02# configure terminal
msk-dasedokhin-gw-02(config)# router ripng
msk-dasedokhin-gw-02(config-router)# network eth0
msk-dasedokhin-gw-02(config-router)# network eth1
msk-dasedokhin-gw-02(config-router)# exit
msk-dasedokhin-gw-02(config)# exit
msk-dasedokhin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-02#
```

Рисунок 29: Настройка RIPv6 для сетей IPv6 на маршрутизаторе gw-02

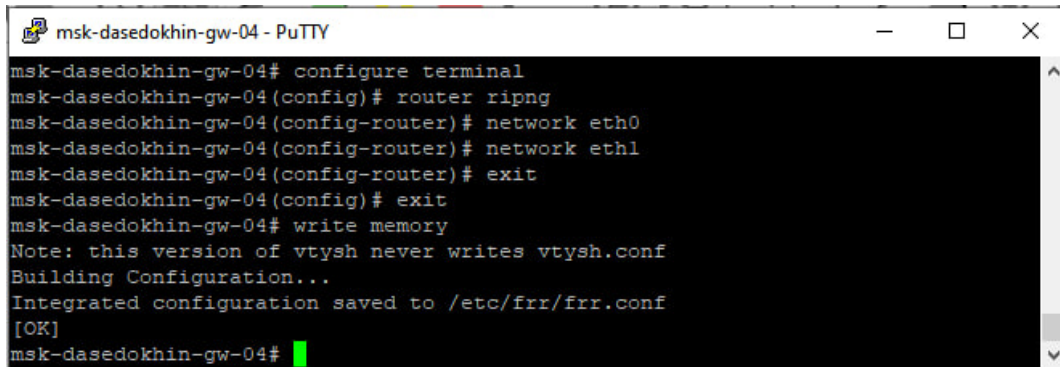
3.30 слайд 30



```
msk-dasedokhin-gw-03# configure terminal
msk-dasedokhin-gw-03(config)# router ripng
msk-dasedokhin-gw-03(config-router)# network eth0
msk-dasedokhin-gw-03(config-router)# network eth1
msk-dasedokhin-gw-03(config-router)# network eth2
msk-dasedokhin-gw-03(config-router)# exit
msk-dasedokhin-gw-03(config)# exit
msk-dasedokhin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-03#
```

Рисунок 30: Настройка RIPv6 для сетей IPv6 на маршрутизаторе gw-03

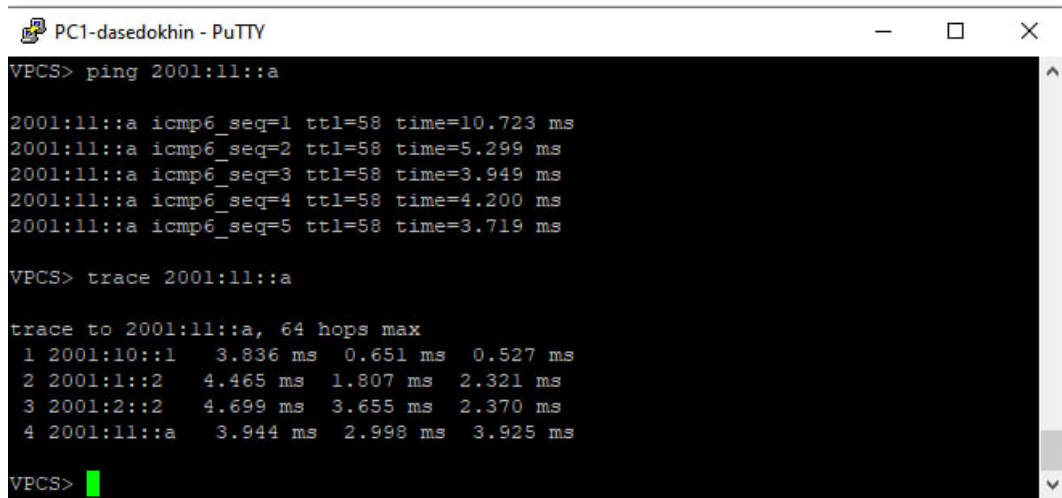
3.31 слайд 31



```
msk-dasedokhin-gw-04# configure terminal
msk-dasedokhin-gw-04(config)# router ripng
msk-dasedokhin-gw-04(config-router)# network eth0
msk-dasedokhin-gw-04(config-router)# network eth1
msk-dasedokhin-gw-04(config-router)# exit
msk-dasedokhin-gw-04(config)# exit
msk-dasedokhin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-04#
```

Рисунок 31: Настройка RIPv6 для сетей IPv6 на маршрутизаторе gw-04

3.32 слайд 32



```
PC1-dasedokhin - PuTTY
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=10.723 ms
2001:11::a icmp6_seq=2 ttl=58 time=5.299 ms
2001:11::a icmp6_seq=3 ttl=58 time=3.949 ms
2001:11::a icmp6_seq=4 ttl=58 time=4.200 ms
2001:11::a icmp6_seq=5 ttl=58 time=3.719 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    3.836 ms  0.651 ms  0.527 ms
 2 2001:1::2     4.465 ms  1.807 ms  2.321 ms
 3 2001:2::2     4.699 ms  3.655 ms  2.370 ms
 4 2001:11::a    3.944 ms  2.998 ms  3.925 ms

VPCS>
```

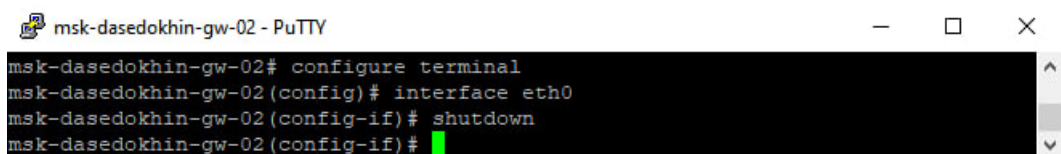
3.33 слайд 33



```
msk-dasedokhin-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

      Network      Next Hop      Via      Metric Tag Time
C(i) 2001:1::/64
      ::          self          1      0
R(n) 2001:2::/64
      fe80::e47:34ff:fe6e:0  eth1        2      0  02:19
R(n) 2001:3::/64
      fe80::e61:ff:fe04:1    eth2        2      0  02:31
C(i) 2001:4::/64
      ::          self          1      0
C(i) 2001:10::/64
      ::          self          1      0
R(n) 2001:11::/64
      ::          self          1      0
```

3.34 слайд 34



```
msk-dasedokhin-gw-02# configure terminal
msk-dasedokhin-gw-02(config)# interface eth0
msk-dasedokhin-gw-02(config-if)# shutdown
msk-dasedokhin-gw-02(config-if)#
```

Рисунок 34: Отключение интерфейса на маршрутизаторе gw-02

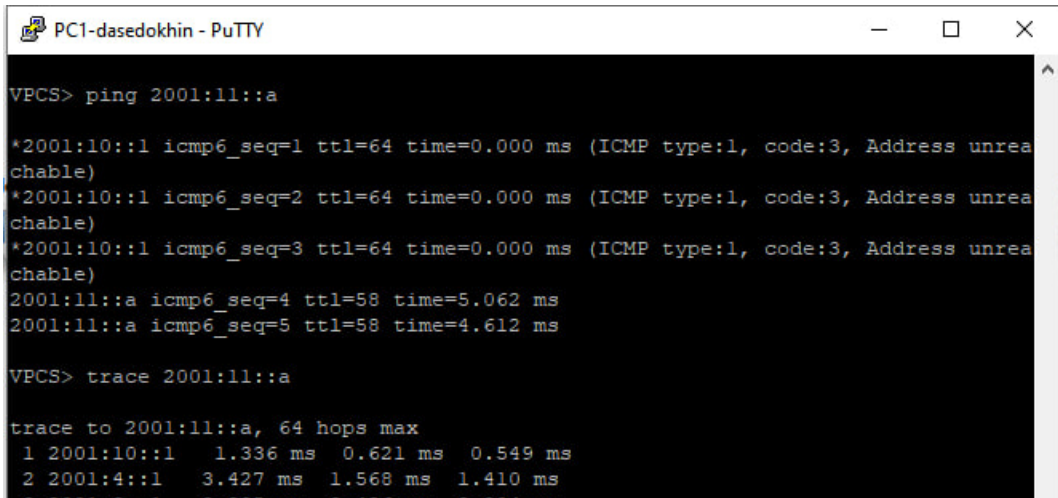
3.35 слайд 35



```
msk-dasedokhin-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

      Network      Next Hop      Via      Metric Tag Time
C(i) 2001:1::/64
      ::          self         1      0
R(n) 2001:2::/64
      fe80::e47:34ff:fe6e:0 eth1        2      0 02:26
R(n) 2001:3::/64
      fe80::e61:ff:fe04:1  eth2        2      0 02:46
C(i) 2001:4::/64
      ::          self         1      0
C(i) 2001:10::/64
      ::          self         1      0
```

3.36 слайд 36



```
PC1-dasedokhin - PuTTY

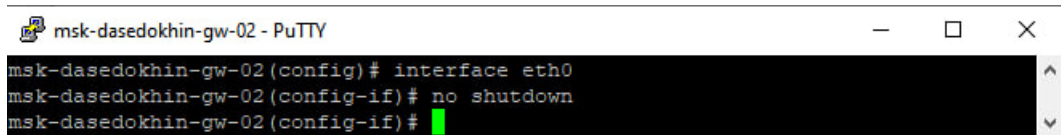
VPCS> ping 2001:11::a

*2001:10::1 icmp6_seq=1 ttl=64 time=0.000 ms (ICMP type:1, code:3, Address unrea
chable)
*2001:10::1 icmp6_seq=2 ttl=64 time=0.000 ms (ICMP type:1, code:3, Address unrea
chable)
*2001:10::1 icmp6_seq=3 ttl=64 time=0.000 ms (ICMP type:1, code:3, Address unrea
chable)
2001:11::a icmp6_seq=4 ttl=58 time=5.062 ms
2001:11::a icmp6_seq=5 ttl=58 time=4.612 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    1.336 ms  0.621 ms  0.549 ms
 2 2001:4::1     3.427 ms  1.568 ms  1.410 ms
```

3.37 слайд 37

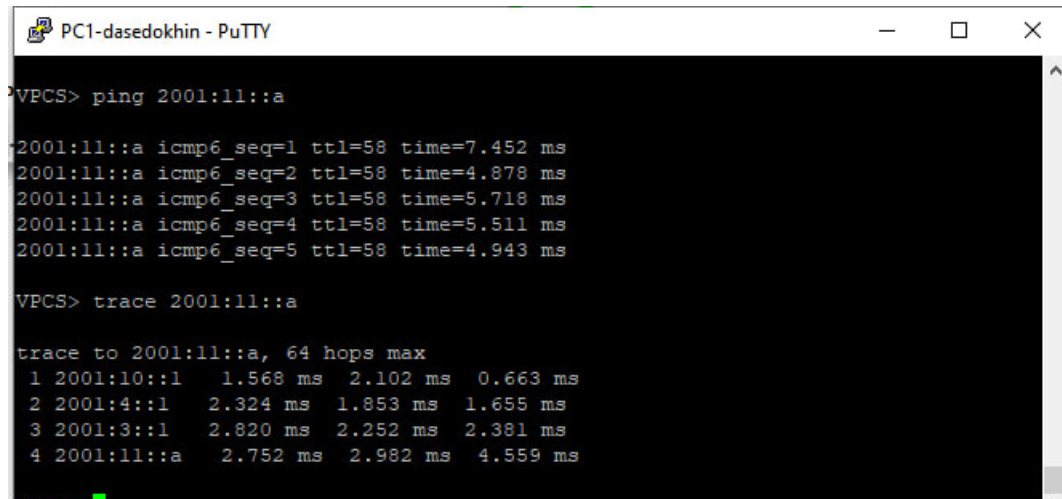


The image shows a PuTTY terminal window titled "msk-dasedokhin-gw-02 - PuTTY". The terminal has a black background with white text. It displays three lines of configuration commands for a router. The first line is "msk-dasedokhin-gw-02 (config)# interface eth0", the second is "msk-dasedokhin-gw-02 (config-if)# no shutdown", and the third is "msk-dasedokhin-gw-02 (config-if)#" followed by a green cursor. The window has standard Windows-style window controls (minimize, maximize, close) in the top right corner.

```
msk-dasedokhin-gw-02 (config)# interface eth0
msk-dasedokhin-gw-02 (config-if)# no shutdown
msk-dasedokhin-gw-02 (config-if)#
```

Рисунок 37: Включение интерфейса на маршрутизаторе gw-02

3.38 слайд 38



```
PC1-dasedokhin - PuTTY

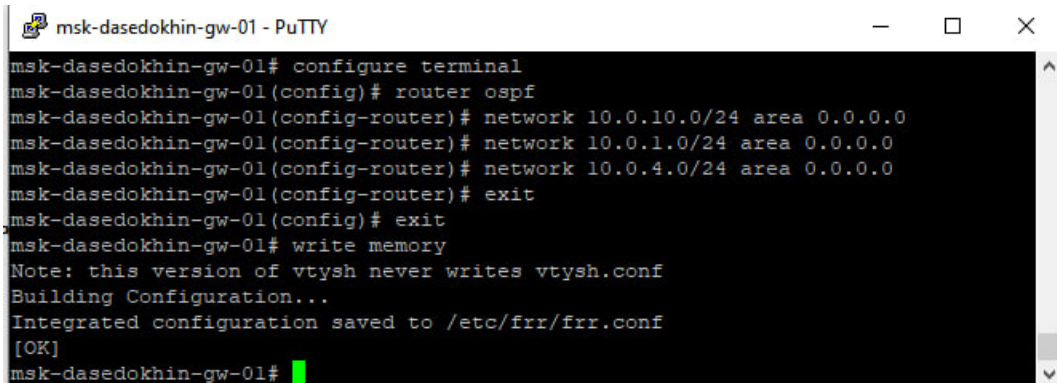
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=7.452 ms
2001:11::a icmp6_seq=2 ttl=58 time=4.878 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.718 ms
2001:11::a icmp6_seq=4 ttl=58 time=5.511 ms
2001:11::a icmp6_seq=5 ttl=58 time=4.943 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1   1.568 ms  2.102 ms  0.663 ms
 2 2001:4::1    2.324 ms  1.853 ms  1.655 ms
 3 2001:3::1    2.820 ms  2.252 ms  2.381 ms
 4 2001:11::a   2.752 ms  2.982 ms  4.559 ms
```

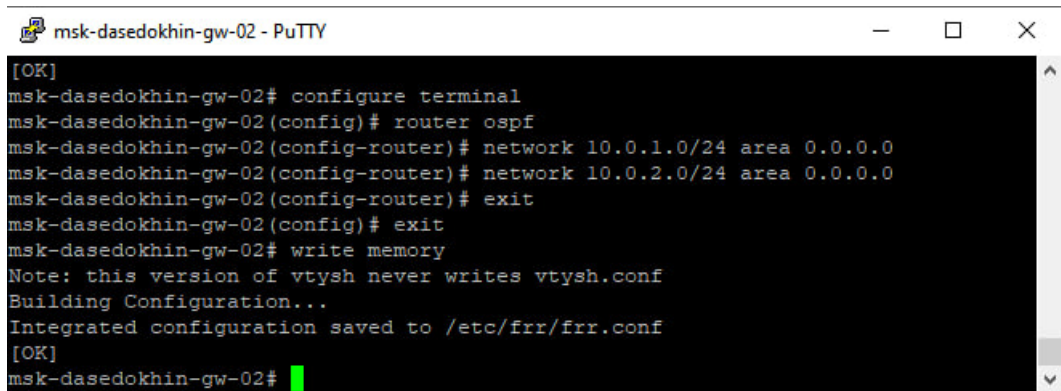
3.39 слайд 39



```
msk-dasedokhin-gw-01# configure terminal
msk-dasedokhin-gw-01(config)# router ospf
msk-dasedokhin-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-dasedokhin-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-dasedokhin-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-dasedokhin-gw-01(config-router)# exit
msk-dasedokhin-gw-01(config)# exit
msk-dasedokhin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-01#
```

Рисунок 39: Настройка OSPFv2 для сетей IPV4 на маршрутизаторе gw-01

3.40 слайд 40

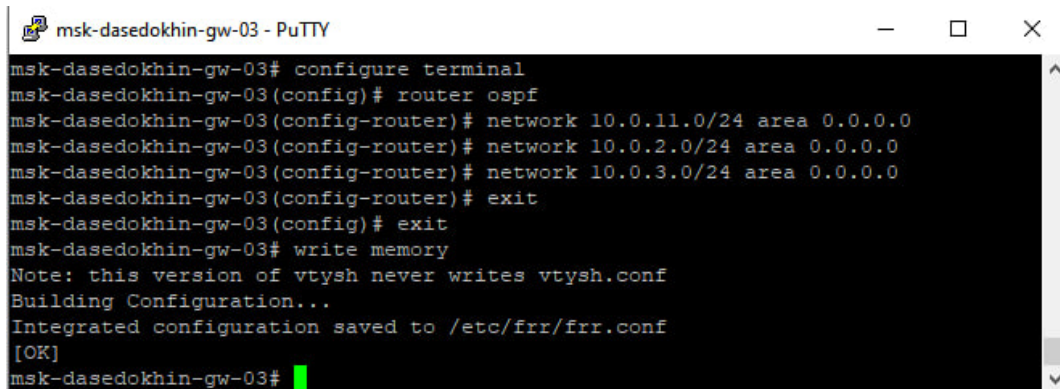


The image shows a PuTTY terminal window titled "msk-dasedokhin-gw-02 - PuTTY". The terminal output displays the configuration of OSPFv2 on a router. The user enters the command "configure terminal" to enter configuration mode, then "router ospf" to enter OSPF configuration mode. Two networks are added: "10.0.1.0/24" and "10.0.2.0/24", both assigned to area "0.0.0.0". After exiting the router configuration mode and then the configuration mode itself, the user enters "write memory" to save the configuration. A message indicates that the configuration is saved to "/etc/frr/frr.conf". The terminal ends with a prompt "msk-dasedokhin-gw-02#" and a green cursor.

```
[OK]
msk-dasedokhin-gw-02# configure terminal
msk-dasedokhin-gw-02(config)# router ospf
msk-dasedokhin-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-dasedokhin-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-dasedokhin-gw-02(config-router)# exit
msk-dasedokhin-gw-02(config)# exit
msk-dasedokhin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-02#
```

Рисунок 40: Настройка OSPFv2 для сетей IPV4 на маршрутизаторе gw-02

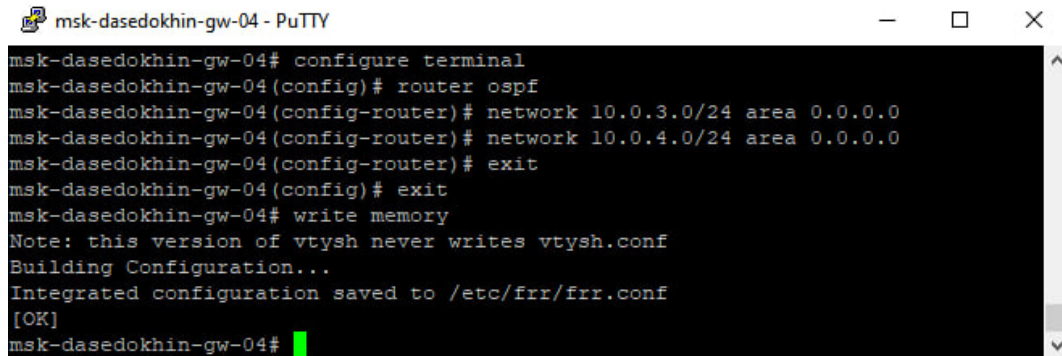
3.41 слайд 41



```
msk-dasedokhin-gw-03# configure terminal
msk-dasedokhin-gw-03(config)# router ospf
msk-dasedokhin-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-dasedokhin-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-dasedokhin-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-dasedokhin-gw-03(config-router)# exit
msk-dasedokhin-gw-03(config)# exit
msk-dasedokhin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-03#
```

Рисунок 41: Настройка OSPFv2 для сетей IPV4 на маршрутизаторе gw-03

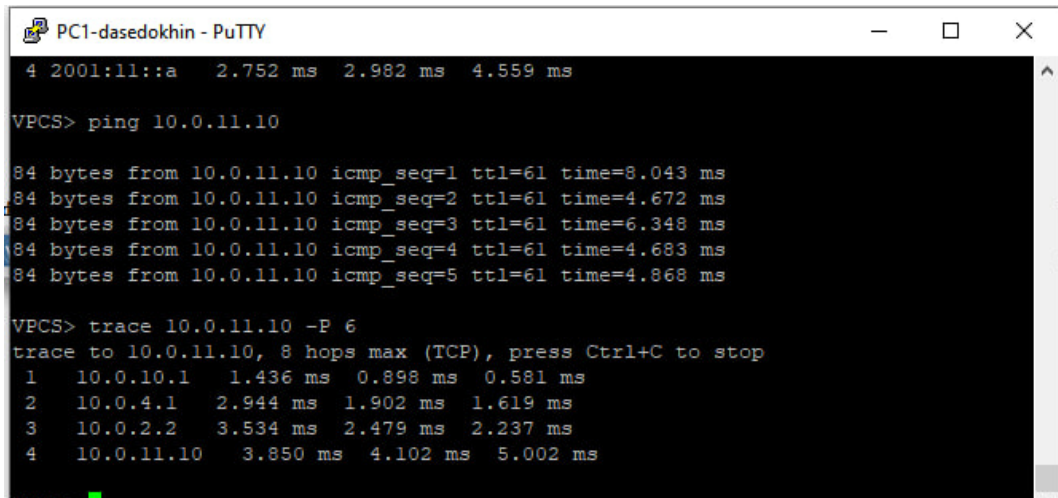
3.42 слайд 42



```
msk-dasedokhin-gw-04# configure terminal
msk-dasedokhin-gw-04(config)# router ospf
msk-dasedokhin-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-dasedokhin-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-dasedokhin-gw-04(config-router)# exit
msk-dasedokhin-gw-04(config)# exit
msk-dasedokhin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-04#
```

Рисунок 42: Настройка OSPFv2 для сетей IPV4 на маршрутизаторе gw-04

3.43 слайд 43



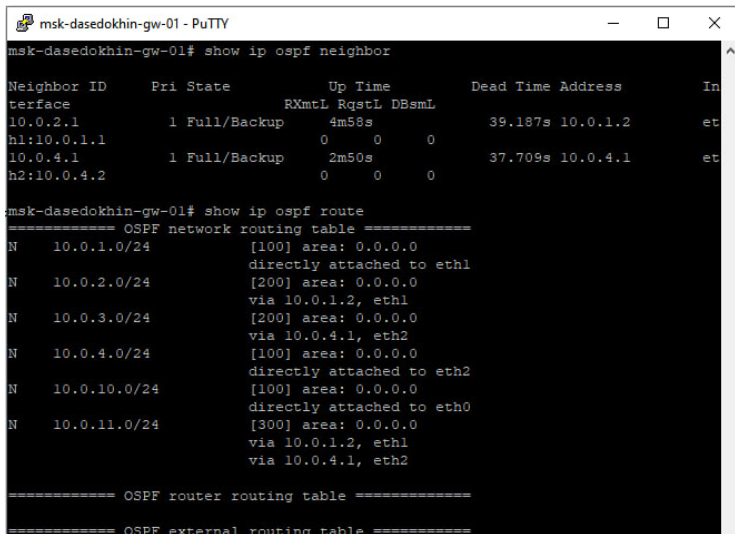
```
PC1-dasedokhin - PuTTY
4 2001:11::a 2.752 ms 2.982 ms 4.559 ms

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=8.043 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=4.672 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=6.348 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=4.683 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=4.868 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1 10.0.10.1 1.436 ms 0.898 ms 0.581 ms
 2 10.0.4.1 2.944 ms 1.902 ms 1.619 ms
 3 10.0.2.2 3.534 ms 2.479 ms 2.237 ms
 4 10.0.11.10 3.850 ms 4.102 ms 5.002 ms
```

3.44 слайд 44



```
msk-dasedokhin-gw-01 - PuTTY
msk-dasedokhin-gw-01# show ip ospf neighbor

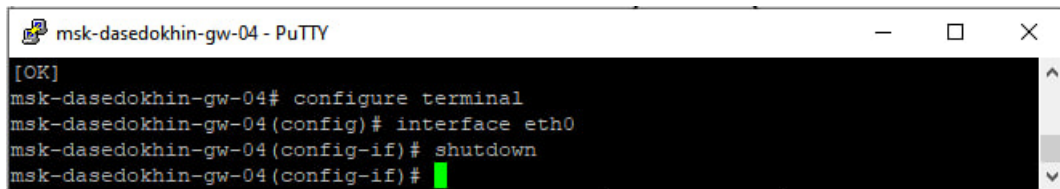
Neighbor ID      Pri State           Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.2.1          1 Full/Backup      4m58s             39.187s 10.0.1.2      et
hl:10.0.1.1       0      0      0
10.0.4.1          1 Full/Backup      2m50s             37.709s 10.0.4.1      et
h2:10.0.4.2       0      0      0

msk-dasedokhin-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24      [100] area: 0.0.0.0
    directly attached to eth1
N   10.0.2.0/24      [200] area: 0.0.0.0
    via 10.0.1.2, eth1
N   10.0.3.0/24      [200] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.4.0/24      [100] area: 0.0.0.0
    directly attached to eth2
N   10.0.10.0/24     [100] area: 0.0.0.0
    directly attached to eth0
N   10.0.11.0/24     [300] area: 0.0.0.0
    via 10.0.1.2, eth1
    via 10.0.4.1, eth2

===== OSPF router routing table =====

===== OSPF external routing table =====
```

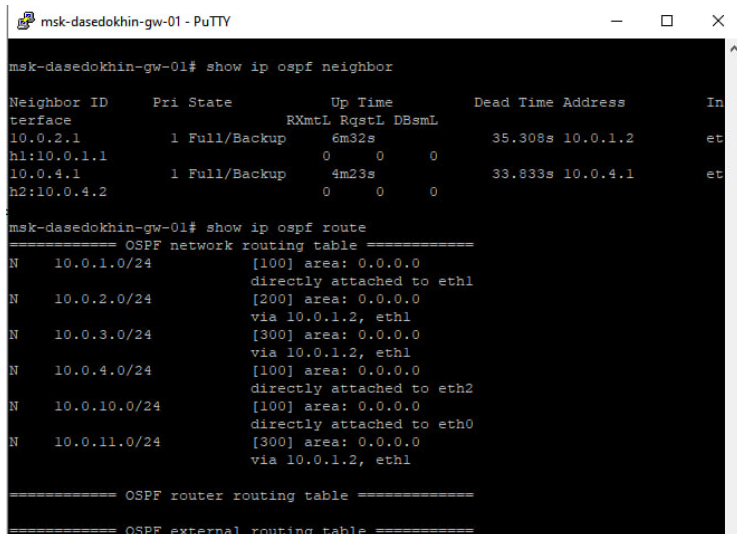
3.45 слайд 45



```
msk-dasedokhin-gw-04 - PuTTY
[OK]
msk-dasedokhin-gw-04# configure terminal
msk-dasedokhin-gw-04(config)# interface eth0
msk-dasedokhin-gw-04(config-if)# shutdown
msk-dasedokhin-gw-04(config-if)#
```

Рисунок 45: Отключение интерфейса на маршрутизаторе gw-04

3.46 слайд 46



```
msk-dasedokhin-gw-01# show ip ospf neighbor
```

Neighbor ID	Pri	State	Up Time	Dead Time	Address	In
terface			RXmtL RqstL DBsmL			
10.0.2.1	1	Full/Backup	6m32s	35.308s	10.0.1.2	et
h1:10.0.1.1			0 0 0			
10.0.4.1	1	Full/Backup	4m23s	33.833s	10.0.4.1	et
h2:10.0.4.2			0 0 0			

```
msk-dasedokhin-gw-01# show ip ospf route
```

```
===== OSPF network routing table =====
```

N	10.0.1.0/24	[100] area: 0.0.0.0
		directly attached to eth1
N	10.0.2.0/24	[200] area: 0.0.0.0
		via 10.0.1.2, eth1
N	10.0.3.0/24	[300] area: 0.0.0.0
		via 10.0.1.2, eth1
N	10.0.4.0/24	[100] area: 0.0.0.0
		directly attached to eth2
N	10.0.10.0/24	[100] area: 0.0.0.0
		directly attached to eth0
N	10.0.11.0/24	[300] area: 0.0.0.0
		via 10.0.1.2, eth1

```
===== OSPF router routing table =====
```

```
===== OSPF external routing table =====
```

3.47 слайд 47

```
PC1-dasedokhin - PuTTY

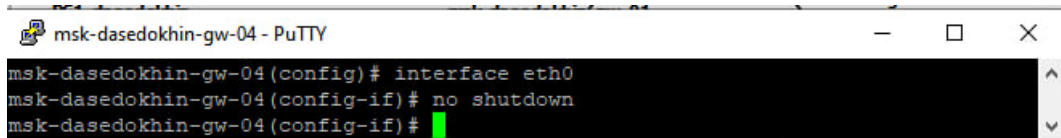
4 10.0.11.10 3.850 ms 4.102 ms 5.002 ms

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=6.147 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=4.987 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=4.244 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=4.361 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=4.701 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1 10.0.10.1 1.542 ms 0.994 ms 0.570 ms
 2 10.0.1.2 2.691 ms 1.604 ms 1.531 ms
 3 10.0.2.2 4.104 ms 3.111 ms 2.389 ms
 4 10.0.11.10 3.058 ms 2.810 ms 3.441 ms
```

3.48 слайд 48



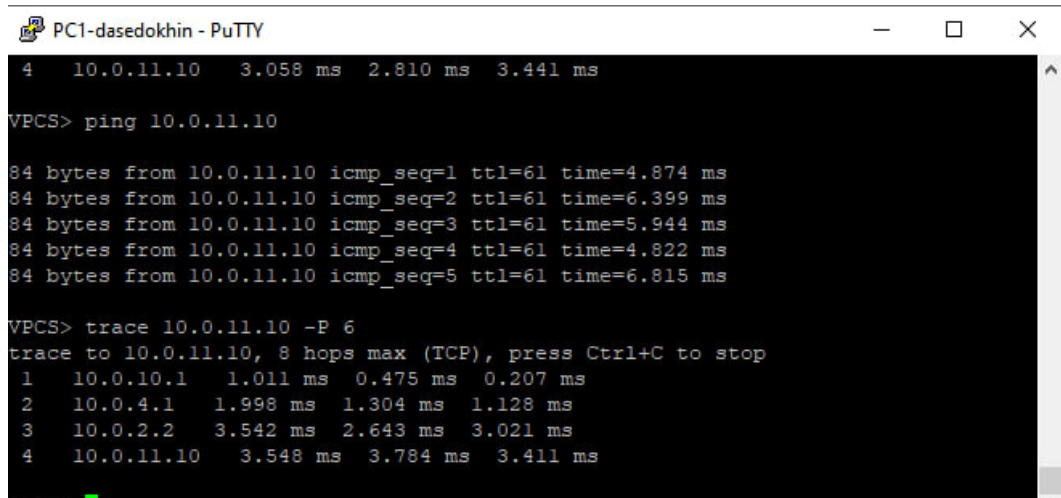
The image shows a PuTTY terminal window titled "msk-dasedokhin-gw-04 - PuTTY". The terminal output is as follows:

```
msk-dasedokhin-gw-04(config)# interface eth0
msk-dasedokhin-gw-04(config-if)# no shutdown
msk-dasedokhin-gw-04(config-if)#
```

The cursor is at the end of the third line, indicated by a green block.

Рисунок 48: Включение интерфейса на маршрутизаторе gw-04

3.49 слайд 49



```
PC1-dasedokhin - PuTTY

4 10.0.11.10 3.058 ms 2.810 ms 3.441 ms

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=4.874 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=6.399 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=5.944 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=4.822 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=6.815 ms

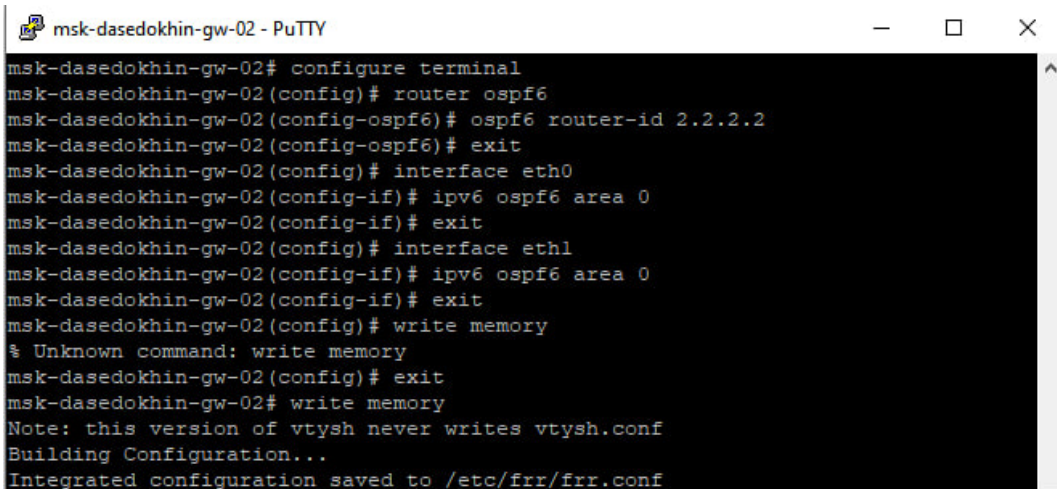
VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1 10.0.10.1 1.011 ms 0.475 ms 0.207 ms
 2 10.0.4.1 1.998 ms 1.304 ms 1.128 ms
 3 10.0.2.2 3.542 ms 2.643 ms 3.021 ms
 4 10.0.11.10 3.548 ms 3.784 ms 3.411 ms
```

3.50 слайд 50

msk-dasedokhin-gw-01 - PuTTY

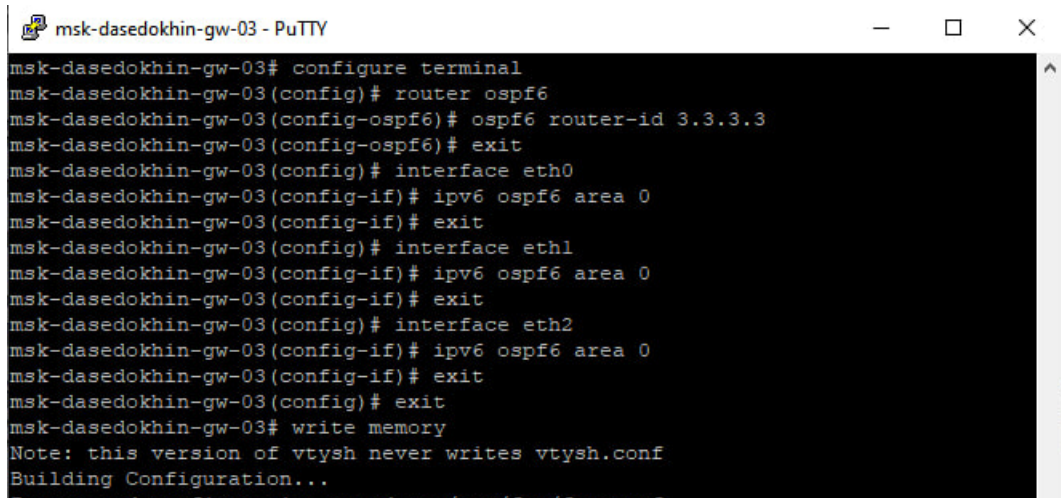
```
msk-dasedokhin-gw-01# configure terminal
msk-dasedokhin-gw-01(config)# router ospf6
msk-dasedokhin-gw-01(config-ospf6)# ospf router-id 1.1.1.1
msk-dasedokhin-gw-01(config-ospf6)# exit
msk-dasedokhin-gw-01(config)# interface eth0
msk-dasedokhin-gw-01(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# interface eth1
msk-dasedokhin-gw-01(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# interface eth2
msk-dasedokhin-gw-01(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# exit
msk-dasedokhin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
```

3.51 слайд 51



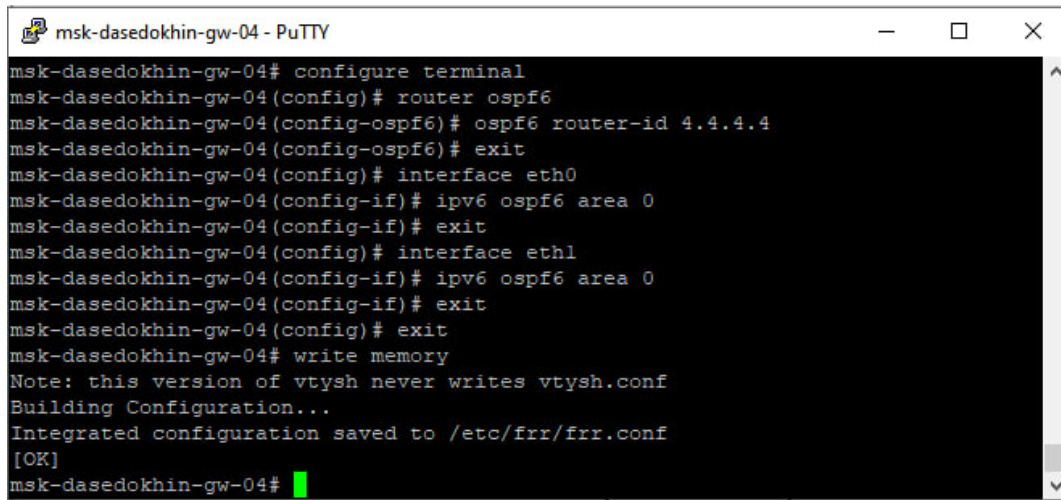
```
msk-dasedokhin-gw-02# configure terminal
msk-dasedokhin-gw-02(config)# router ospf6
msk-dasedokhin-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-dasedokhin-gw-02(config-ospf6)# exit
msk-dasedokhin-gw-02(config)# interface eth0
msk-dasedokhin-gw-02(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-02(config-if)# exit
msk-dasedokhin-gw-02(config)# interface eth1
msk-dasedokhin-gw-02(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-02(config-if)# exit
msk-dasedokhin-gw-02(config)# write memory
% Unknown command: write memory
msk-dasedokhin-gw-02(config)# exit
msk-dasedokhin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
```

3.52 слайд 52



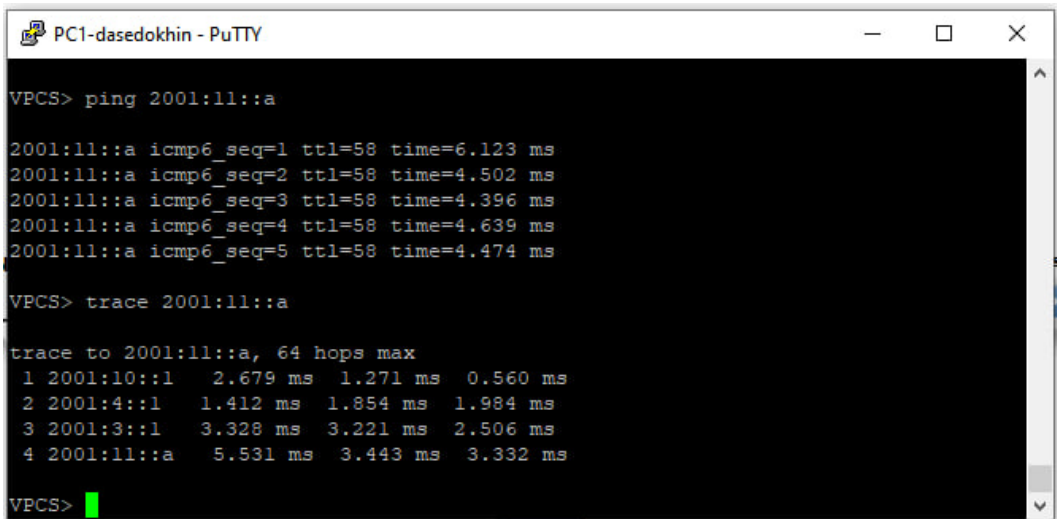
```
msk-dasedokhin-gw-03# configure terminal
msk-dasedokhin-gw-03(config)# router ospf6
msk-dasedokhin-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-dasedokhin-gw-03(config-ospf6)# exit
msk-dasedokhin-gw-03(config)# interface eth0
msk-dasedokhin-gw-03(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-03(config-if)# exit
msk-dasedokhin-gw-03(config)# interface eth1
msk-dasedokhin-gw-03(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-03(config-if)# exit
msk-dasedokhin-gw-03(config)# interface eth2
msk-dasedokhin-gw-03(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-03(config-if)# exit
msk-dasedokhin-gw-03(config)# exit
msk-dasedokhin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
```

3.53 слайд 53



```
msk-dasedokhin-gw-04# configure terminal
msk-dasedokhin-gw-04(config)# router ospf6
msk-dasedokhin-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-dasedokhin-gw-04(config-ospf6)# exit
msk-dasedokhin-gw-04(config)# interface eth0
msk-dasedokhin-gw-04(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-04(config-if)# exit
msk-dasedokhin-gw-04(config)# interface eth1
msk-dasedokhin-gw-04(config-if)# ipv6 ospf6 area 0
msk-dasedokhin-gw-04(config-if)# exit
msk-dasedokhin-gw-04(config)# exit
msk-dasedokhin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-04#
```

3.54 слайд 54



```
PC1-dasedokhin - PuTTY

VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=6.123 ms
2001:11::a icmp6_seq=2 ttl=58 time=4.502 ms
2001:11::a icmp6_seq=3 ttl=58 time=4.396 ms
2001:11::a icmp6_seq=4 ttl=58 time=4.639 ms
2001:11::a icmp6_seq=5 ttl=58 time=4.474 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    2.679 ms  1.271 ms  0.560 ms
 2 2001:4::1    1.412 ms  1.854 ms  1.984 ms
 3 2001:3::1    3.328 ms  3.221 ms  2.506 ms
 4 2001:11::a   5.531 ms  3.443 ms  3.332 ms

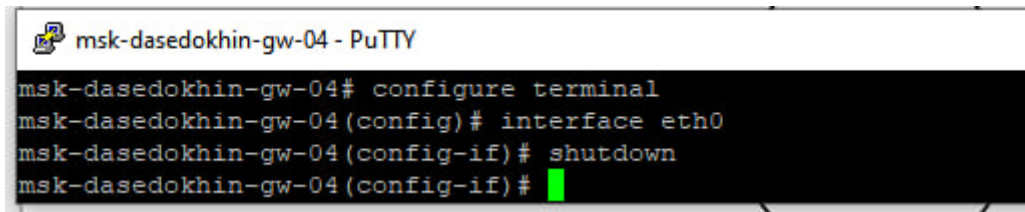
VPCS> 
```

3.55 слайд 55

```
msk-dasedokhin-gw-01# show ipv6 ospf6 neighbor
Neighbor ID      Pri      DeadTime      State/IfState      Duration I/F[State]
2.2.2.2          1        00:00:31      Full/BDR          00:10:56 eth1[DR]
4.4.4.4          1        00:00:38      Full/BDR          00:00:56 eth2[DR]
msk-dasedokhin-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64      ::              eth1 00:01:11
*N IA 2001:2::/64      fe80::e47:34ff:fe6e:0 eth1 00:01:11
*N IA 2001:3::/64      fe80::e61:ff:fe04:1  eth2 00:01:11
*N IA 2001:4::/64      ::              eth2 00:01:11
*N IA 2001:10::/64     ::              eth0 00:01:11
*N IA 2001:11::/64     fe80::e47:34ff:fe6e:0 eth1 00:01:11
                       fe80::e61:ff:fe04:1  eth2
```

Рисунок 55: Проверка таблицы маршрутизации протокола OSPFv3

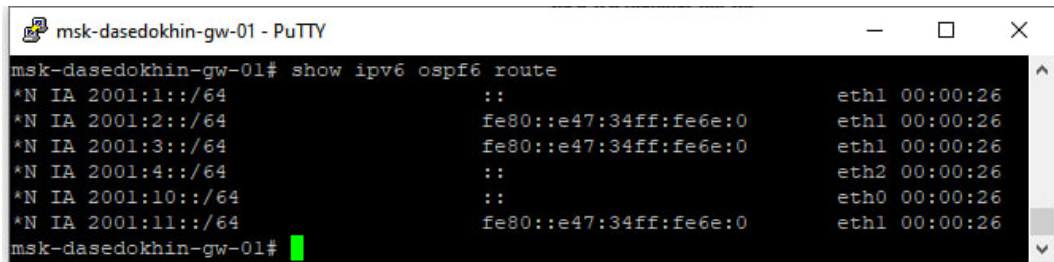
3.56 слайд 56



```
msk-dasedokhin-gw-04# configure terminal
msk-dasedokhin-gw-04(config)# interface eth0
msk-dasedokhin-gw-04(config-if)# shutdown
msk-dasedokhin-gw-04(config-if)#
```

Рисунок 56: Отключение интерфейса на маршрутизаторе gw-04

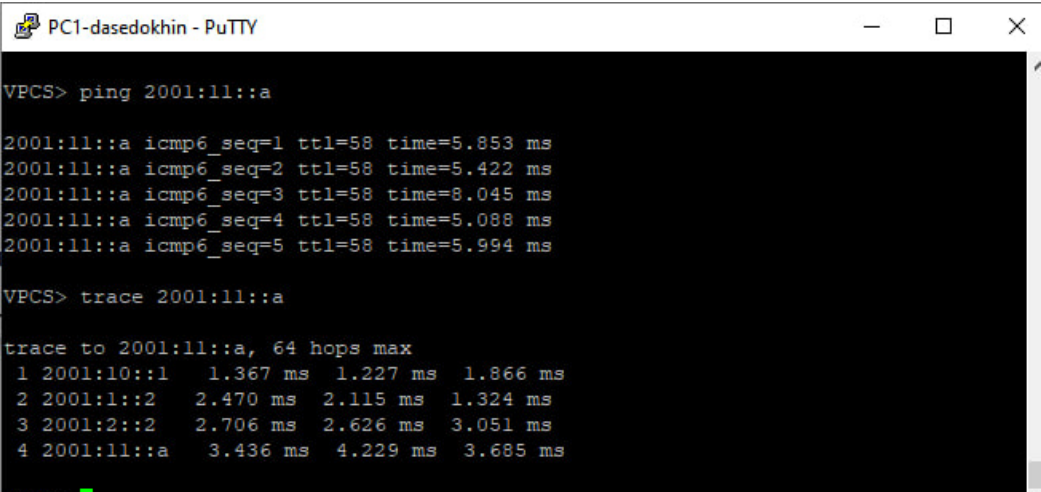
3.57 слайд 57



```
msk-dasedokhin-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64          ::          eth1 00:00:26
*N IA 2001:2::/64          fe80::e47:34ff:fe6e:0 eth1 00:00:26
*N IA 2001:3::/64          fe80::e47:34ff:fe6e:0 eth1 00:00:26
*N IA 2001:4::/64          ::          eth2 00:00:26
*N IA 2001:10::/64         ::          eth0 00:00:26
*N IA 2001:11::/64         fe80::e47:34ff:fe6e:0 eth1 00:00:26
msk-dasedokhin-gw-01#
```

Рисунок 57: Проверка таблицы маршрутизации протокола OSPFv3

3.58 слайд 58



```
PC1-dasedokhin - PuTTY

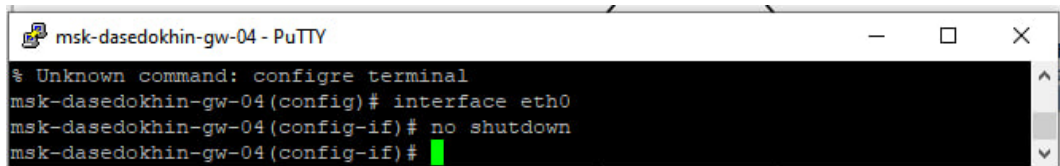
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=5.853 ms
2001:11::a icmp6_seq=2 ttl=58 time=5.422 ms
2001:11::a icmp6_seq=3 ttl=58 time=8.045 ms
2001:11::a icmp6_seq=4 ttl=58 time=5.088 ms
2001:11::a icmp6_seq=5 ttl=58 time=5.994 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1   1.367 ms  1.227 ms  1.866 ms
 2 2001:1::2    2.470 ms  2.115 ms  1.324 ms
 3 2001:2::2    2.706 ms  2.626 ms  3.051 ms
 4 2001:11::a   3.436 ms  4.229 ms  3.685 ms
```

3.59 слайд 59



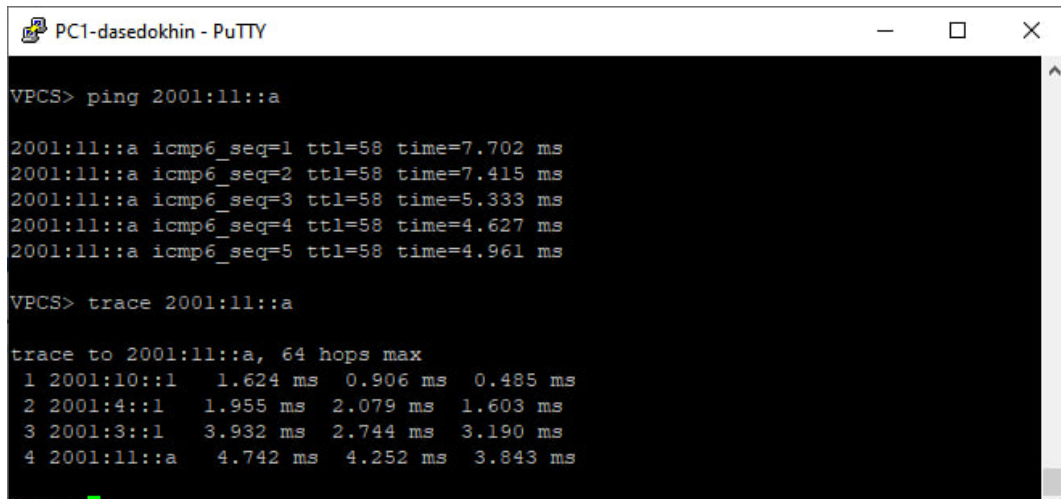
The image shows a PuTTY terminal window titled "msk-dasedokhin-gw-04 - PuTTY". The terminal output is as follows:

```
% Unknown command: configre terminal
msk-dasedokhin-gw-04(config)# interface eth0
msk-dasedokhin-gw-04(config-if)# no shutdown
msk-dasedokhin-gw-04(config-if)#
```

The last line shows a green cursor, indicating the prompt is ready for the next command.

Рисунок 59: Включение интерфейса на маршрутизаторе gw-04

3.60 слайд 60



```
PC1-dasedokhin - PuTTY

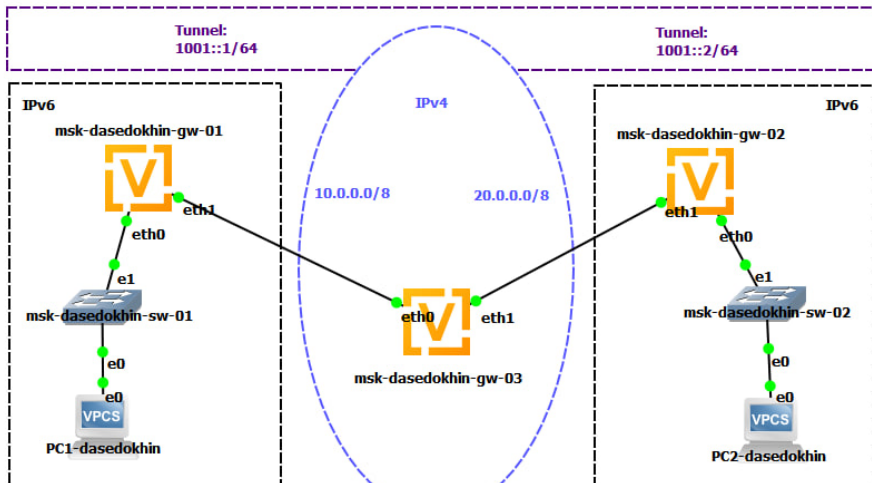
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=7.702 ms
2001:11::a icmp6_seq=2 ttl=58 time=7.415 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.333 ms
2001:11::a icmp6_seq=4 ttl=58 time=4.627 ms
2001:11::a icmp6_seq=5 ttl=58 time=4.961 ms

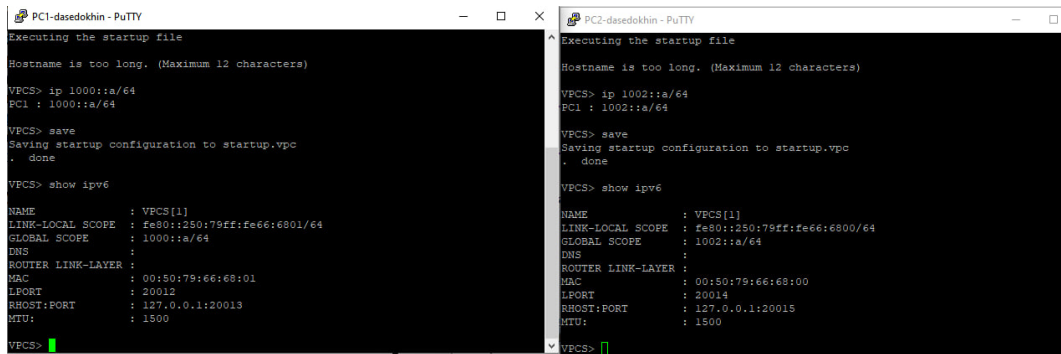
VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    1.624 ms  0.906 ms  0.485 ms
 2 2001:4::1    1.955 ms  2.079 ms  1.603 ms
 3 2001:3::1    3.932 ms  2.744 ms  3.190 ms
 4 2001:11::a   4.742 ms  4.252 ms  3.843 ms
```

3.61 слайд 61



3.62 слайд 62



```
PC1-dasedokhin - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 1000::a/64
PC1 : 1000::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE     : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE         : 1000::a/64
DNS                  :
ROUTER LINK-LAYER    :
MAC                  : 00:50:79:66:68:01
LPORT                : 20012
RHOST:PORT           : 127.0.0.1:20013
MTU                  : 1500

VPCS>
```

```
PC2-dasedokhin - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 1002::a/64
PC1 : 1002::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

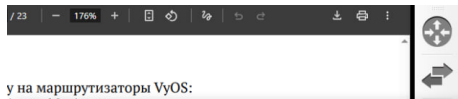
VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE     : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE         : 1002::a/64
DNS                  :
ROUTER LINK-LAYER    :
MAC                  : 00:50:79:66:68:00
LPORT                : 20014
RHOST:PORT           : 127.0.0.1:20015
MTU                  : 1500

VPCS>
```

Рисунок 62: Присвоение адресов оконченным устройствам PC1 и PC2

3.63 слайд 63



```

Welcome to VyOS - vyos ttyS0

vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$

```

устройства):
 nfigure
 t system host-name msk-user-gw-0x
 mpare
 mmit
 ve
 it
 boot
 из интерфейсах маршрутизаторов:

```

msk-dasedokhin-gw-02 - PuTTY

[ OK ] Listening on ACPI Listen Socket.
[ OK ] Reached target Sockets.
[ OK ] Started Daily rotation of log files.
[ OK ] Reached target Timers.
[ OK ] Found device /dev/ttyS0.
[ OK ] Started live-config contaoot process (late userspace)..
[ OK ] Reached target Basic System.
Starting OpenBSD Secure Shell session cleanup...
[ OK ] Started Regular background program processing daemon.
Starting Deferred execution scheduler...
Starting Login Service...
Starting LSB: Brings up/down network automatically...
[ OK ] Started D-Bus System Message Bus.
Starting Podman auto-update service...
[ OK ] Started VyOS Router.
[ OK ] Started ACPI event daemon.
Starting System Logging Service...
Starting Atop process accounting daemon...
Starting Permit User Sessions...
[ OK ] Started OpenBSD Secure Shell session cleanup.
[ OK ] Started Deferred execution scheduler.
[ 290.604925] vyos-router[898]: Waiting for NICs to settle down: settled in 0s
c..

```

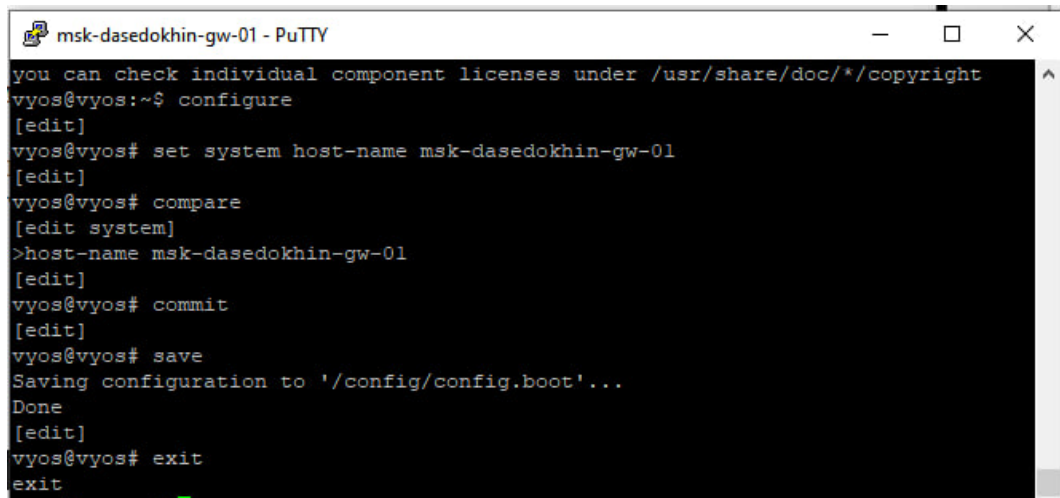
```

msk-dasedokhin-gw-03 - PuTTY

[ OK ] Reached target Local File Systems.
Starting Flush Journal to Persistent Storage...
[ OK ] Started Flush Journal to Persistent Storage.
Starting Create Volatile Files and Directories...
[ OK ] Started udev Kernel Device Manager.
[ OK ] Started Create Volatile Files and Directories.
[ OK ] Reached target System Time Synchronized.
[ OK ] Started Entropy daemon using the HAVEGE algorithm.
Starting live-config conta.t process (late userspace)....
Starting Update UTMP about System Boot/Shutdown...
[ OK ] Started Update UTMP about System Boot/Shutdown.
[ OK ] Reached target System Initialization.
[ OK ] Listening on Podman API Socket.
[ OK ] Started Daily rotation of log files.
[ OK ] Started Podman auto-update timer.
[ OK ] Listening on D-Bus System Message Bus Socket.
[ OK ] Listening on UID daemon activation socket.
[ OK ] Listening on ACPI Listen Socket.

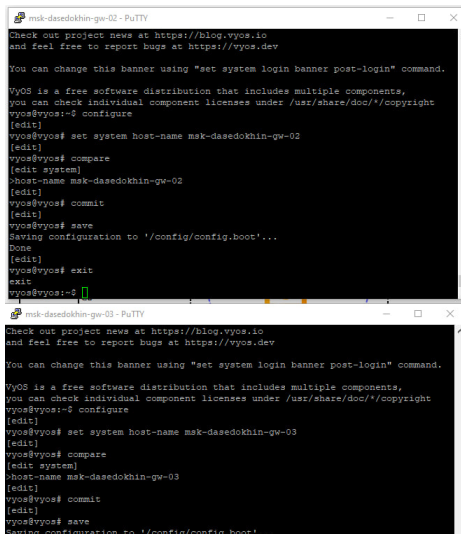
```

3.64 слайд 64



```
msk-dasedokhin-gw-01 - PuTTY
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dasedokhin-gw-01
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dasedokhin-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
```

3.65 слайд 65



The image displays two terminal windows from a PuTTY application, both titled 'msk-dasedokhin-gw-02 - PuTTY' and 'msk-dasedokhin-gw-03 - PuTTY'. Each window shows a VyOS command-line interface. The top window (gw-02) shows the user entering 'configure', then 'set system host-name msk-dasedokhin-gw-02', followed by 'compare', 'commit', and 'save'. The bottom window (gw-03) shows the same sequence of commands, but the 'compare' command is not yet executed, as the prompt is still in the configuration mode. Both windows have a black background with white text. The bottom window's text is partially cut off at the bottom.

```
msk-dasedokhin-gw-02 - PuTTY
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

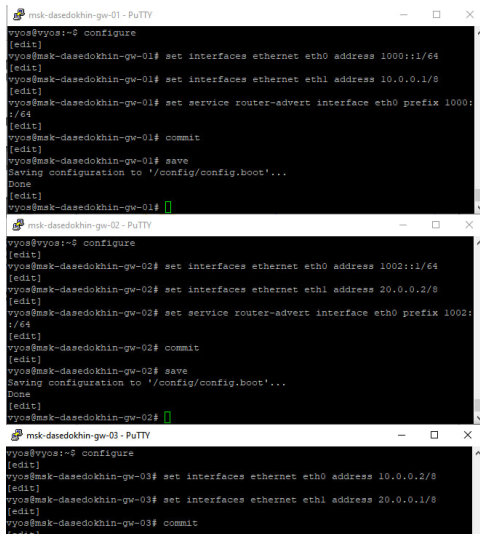
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dasedokhin-gw-02
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dasedokhin-gw-02
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$

msk-dasedokhin-gw-03 - PuTTY
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dasedokhin-gw-03
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dasedokhin-gw-03
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'
```

3.66 слайд 66



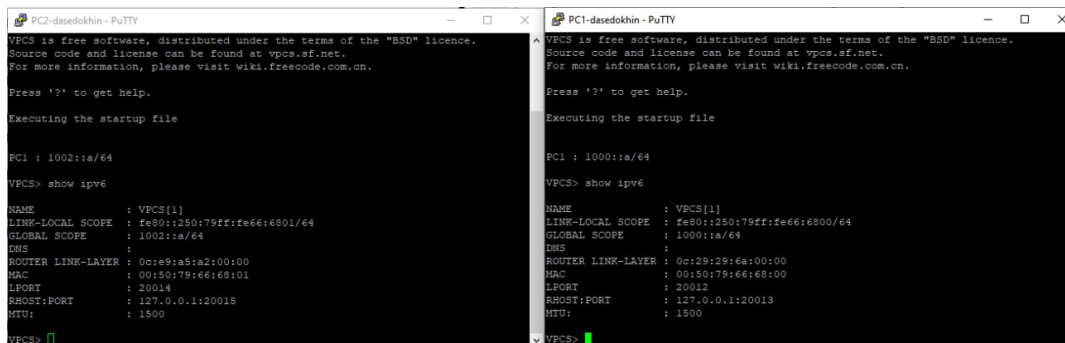
The image displays three terminal windows, each representing a VyOS gateway configuration. The windows are titled 'msk-dasedokhin-gw-01 - PuTTY', 'msk-dasedokhin-gw-02 - PuTTY', and 'msk-dasedokhin-gw-03 - PuTTY'. Each terminal shows the following sequence of commands and output:

```
vyos@vyos:~$ configure
[edit]
vyos@msk-dasedokhin-gw-01# set interfaces ethernet eth0 address 1000::1/64
[edit]
vyos@msk-dasedokhin-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
[edit]
vyos@msk-dasedokhin-gw-01# set service router-advert interface eth0 prefix 1000::/64
[edit]
vyos@msk-dasedokhin-gw-01# commit
[edit]
vyos@msk-dasedokhin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dasedokhin-gw-01#
```

```
vyos@vyos:~$ configure
[edit]
vyos@msk-dasedokhin-gw-02# set interfaces ethernet eth0 address 1002::1/64
[edit]
vyos@msk-dasedokhin-gw-02# set interfaces ethernet eth1 address 20.0.0.2/8
[edit]
vyos@msk-dasedokhin-gw-02# set service router-advert interface eth0 prefix 1002::/64
[edit]
vyos@msk-dasedokhin-gw-02# commit
[edit]
vyos@msk-dasedokhin-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dasedokhin-gw-02#
```

```
vyos@vyos:~$ configure
[edit]
vyos@msk-dasedokhin-gw-03# set interfaces ethernet eth0 address 10.0.0.2/8
[edit]
vyos@msk-dasedokhin-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8
[edit]
vyos@msk-dasedokhin-gw-03# commit
[edit]
```

3.67 слайд 67



The image shows two side-by-side PuTTY terminal windows. The left window is titled 'PC2-dasedokhin - PuTTY' and the right window is titled 'PC1-dasedokhin - PuTTY'. Both windows display the same text: 'VPCS is free software, distributed under the terms of the "BSD" licence. Source code and license can be found at vpcs.sf.net. For more information, please visit wiki.freecode.com.cn. Press '?' to get help. Executing the startup file'. In the left window, the command 'PC1 : 1002::a/64' is entered, followed by 'VPCS> show ipv6', which displays the following output: NAME : VPCS[1], LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6801/64, GLOBAL SCOPE : 1002::a/64, DNS : , ROUTER LINK-LAYER : 0c:e9:a5:a2:00:00, MAC : 00:50:79:66:68:01, LPORT : 20014, RHOST:PORT : 127.0.0.1:20015, MTU: : 1500. In the right window, the command 'PC1 : 1000::a/64' is entered, followed by 'VPCS> show ipv6', which displays the following output: NAME : VPCS[1], LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64, GLOBAL SCOPE : 1000::a/64, DNS : , ROUTER LINK-LAYER : 0c:29:29:6a:00:00, MAC : 00:50:79:66:68:00, LPORT : 20012, RHOST:PORT : 127.0.0.1:20013, MTU: : 1500.

```
PC2-dasedokhin - PuTTY
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1 : 1002::a/64

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE    : 1002::a/64
DNS             :
ROUTER LINK-LAYER : 0c:e9:a5:a2:00:00
MAC             : 00:50:79:66:68:01
LPORT           : 20014
RHOST:PORT      : 127.0.0.1:20015
MTU:            : 1500

VPCS>

PC1-dasedokhin - PuTTY
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1 : 1000::a/64

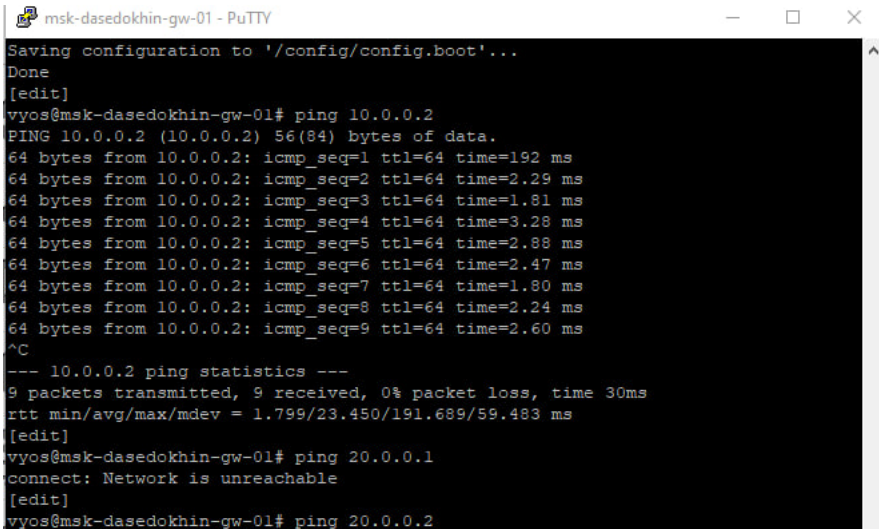
VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 1000::a/64
DNS             :
ROUTER LINK-LAYER : 0c:29:29:6a:00:00
MAC             : 00:50:79:66:68:00
LPORT           : 20012
RHOST:PORT      : 127.0.0.1:20013
MTU:            : 1500

VPCS>
```

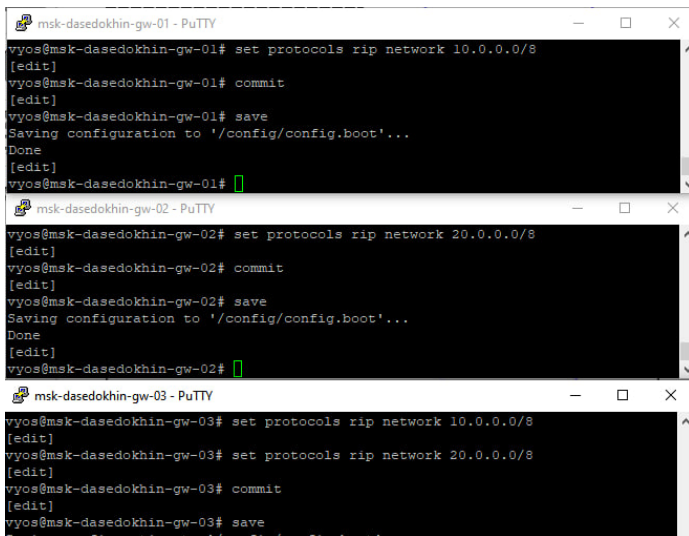
Рисунок 67: Проверка адресов ближайших маршрутизаторов к конечным устройствам

3.68 слайд 68



```
msk-dasedokhin-gw-01 - PuTTY
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dasedokhin-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=192 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=2.29 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=1.81 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=3.28 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=2.88 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=2.47 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=1.80 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=2.24 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=2.60 ms
^C
--- 10.0.0.2 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 30ms
rtt min/avg/max/mdev = 1.799/23.450/191.689/59.483 ms
[edit]
vyos@msk-dasedokhin-gw-01# ping 20.0.0.1
connect: Network is unreachable
[edit]
vyos@msk-dasedokhin-gw-01# ping 20.0.0.2
```

3.69 слайд 69



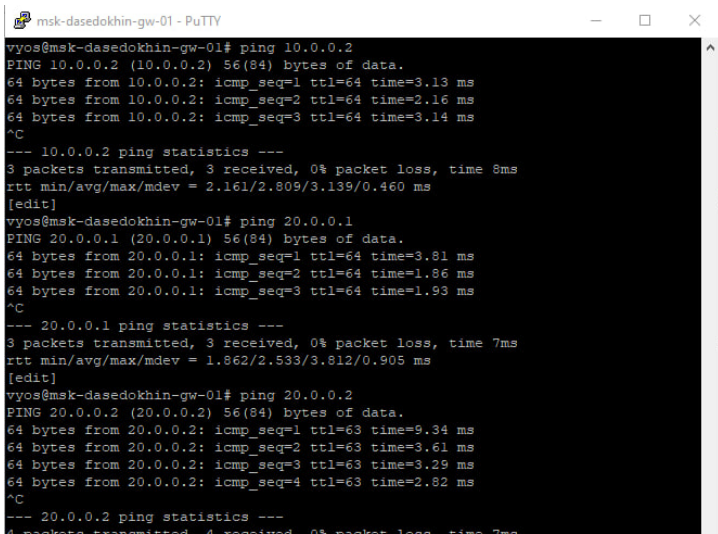
The image displays three terminal windows, each representing a different VyOS router in a network topology. The windows are titled 'msk-dasedokhin-gw-01 - PuTTY', 'msk-dasedokhin-gw-02 - PuTTY', and 'msk-dasedokhin-gw-03 - PuTTY'. Each window shows a sequence of commands used to configure the routers. In the first window, the 'set protocols rip network 10.0.0.0/8' command is entered, followed by '[edit]', 'commit', '[edit]', 'save', and a confirmation message 'Saving configuration to '/config/config.boot'...' Done. The second window shows the same sequence of commands but for the network 20.0.0.0/8. The third window shows the configuration for both networks, 10.0.0.0/8 and 20.0.0.0/8, followed by the same sequence of confirmation commands. The prompt 'vyos@msk-dasedokhin-gw-03#' is visible at the bottom of the third window, indicating the configuration process is still ongoing.

```
msk-dasedokhin-gw-01 - PuTTY
vyos@msk-dasedokhin-gw-01# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-dasedokhin-gw-01# commit
[edit]
vyos@msk-dasedokhin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dasedokhin-gw-01#

msk-dasedokhin-gw-02 - PuTTY
vyos@msk-dasedokhin-gw-02# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-dasedokhin-gw-02# commit
[edit]
vyos@msk-dasedokhin-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dasedokhin-gw-02#

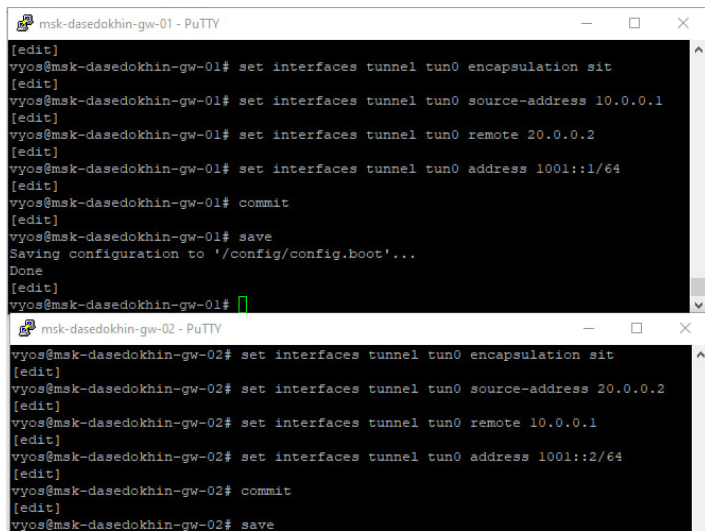
msk-dasedokhin-gw-03 - PuTTY
vyos@msk-dasedokhin-gw-03# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-dasedokhin-gw-03# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-dasedokhin-gw-03# commit
[edit]
vyos@msk-dasedokhin-gw-03# save
```

3.70 слайд 70



```
msk-dasedokhin-gw-01 - PuTTY
vyos@msk-dasedokhin-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.13 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=2.16 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=3.14 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 8ms
rtt min/avg/max/mdev = 2.161/2.809/3.139/0.460 ms
[edit]
vyos@msk-dasedokhin-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=3.81 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=1.86 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=1.93 ms
^C
--- 20.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 7ms
rtt min/avg/max/mdev = 1.862/2.533/3.812/0.905 ms
[edit]
vyos@msk-dasedokhin-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=9.34 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=3.61 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=3.29 ms
64 bytes from 20.0.0.2: icmp_seq=4 ttl=63 time=2.82 ms
^C
--- 20.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 7ms
```

3.71 слайд 71

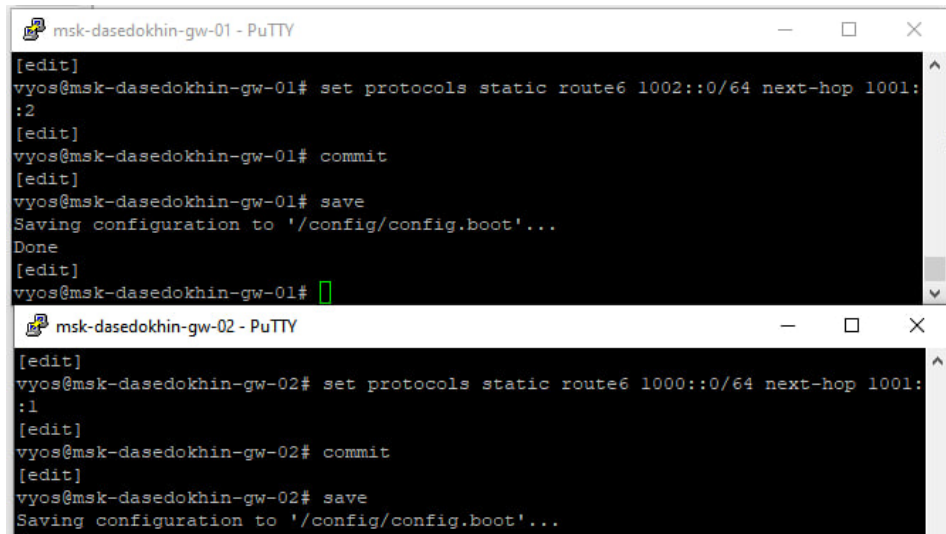


The image shows two terminal windows from a PuTTY application. The top window is titled 'msk-dasedokhin-gw-01 - PuTTY' and shows the configuration of a tunnel interface 'tun0' on router 'msk-dasedokhin-gw-01'. The bottom window is titled 'msk-dasedokhin-gw-02 - PuTTY' and shows the configuration of a tunnel interface 'tun0' on router 'msk-dasedokhin-gw-02'.

```
msk-dasedokhin-gw-01 - PuTTY
[edit]
vyos@msk-dasedokhin-gw-01# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-dasedokhin-gw-01# set interfaces tunnel tun0 source-address 10.0.0.1
[edit]
vyos@msk-dasedokhin-gw-01# set interfaces tunnel tun0 remote 20.0.0.2
[edit]
vyos@msk-dasedokhin-gw-01# set interfaces tunnel tun0 address 1001::1/64
[edit]
vyos@msk-dasedokhin-gw-01# commit
[edit]
vyos@msk-dasedokhin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dasedokhin-gw-01#

msk-dasedokhin-gw-02 - PuTTY
vyos@msk-dasedokhin-gw-02# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-dasedokhin-gw-02# set interfaces tunnel tun0 source-address 20.0.0.2
[edit]
vyos@msk-dasedokhin-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-dasedokhin-gw-02# set interfaces tunnel tun0 address 1001::2/64
[edit]
vyos@msk-dasedokhin-gw-02# commit
[edit]
vyos@msk-dasedokhin-gw-02# save
```

3.72 слайд 72



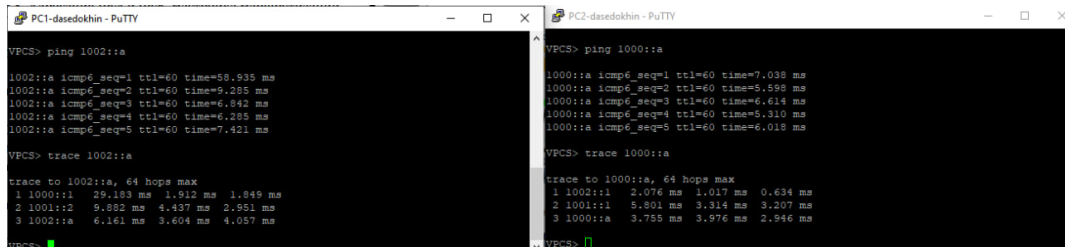
The image shows two PuTTY terminal windows. The top window, titled 'msk-dasedokhin-gw-01 - PuTTY', displays the following commands and output:

```
[edit]
vyos@msk-dasedokhin-gw-01# set protocols static route6 1002::0/64 next-hop 1001:
:2
[edit]
vyos@msk-dasedokhin-gw-01# commit
[edit]
vyos@msk-dasedokhin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dasedokhin-gw-01#
```

The bottom window, titled 'msk-dasedokhin-gw-02 - PuTTY', displays the following commands and output:

```
[edit]
vyos@msk-dasedokhin-gw-02# set protocols static route6 1000::0/64 next-hop 1001:
:1
[edit]
vyos@msk-dasedokhin-gw-02# commit
[edit]
vyos@msk-dasedokhin-gw-02# save
Saving configuration to '/config/config.boot'...
```

3.73 слайд 73



The image shows two terminal windows side-by-side, both titled "PC1-dasedokhin - PuTTY".

Left Terminal (PC1):

```
VPCS> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=58.935 ms
1002::a icmp6_seq=2 ttl=60 time=9.285 ms
1002::a icmp6_seq=3 ttl=60 time=6.842 ms
1002::a icmp6_seq=4 ttl=60 time=6.285 ms
1002::a icmp6_seq=5 ttl=60 time=7.421 ms

VPCS> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1  29.183 ms  1.912 ms  1.849 ms
 2 1001::2   9.882 ms  4.437 ms  2.951 ms
 3 1002::a   6.161 ms  3.604 ms  4.057 ms

VPCS>
```

Right Terminal (PC2):

```
VPCS> ping 1000::a

1000::a icmp6_seq=1 ttl=60 time=7.038 ms
1000::a icmp6_seq=2 ttl=60 time=5.598 ms
1000::a icmp6_seq=3 ttl=60 time=6.614 ms
1000::a icmp6_seq=4 ttl=60 time=5.310 ms
1000::a icmp6_seq=5 ttl=60 time=6.018 ms

VPCS> trace 1000::a

trace to 1000::a, 64 hops max
 1 1002::1   2.076 ms  1.017 ms  0.634 ms
 2 1001::1   5.801 ms  3.314 ms  3.207 ms
 3 1000::a   3.755 ms  3.976 ms  2.946 ms

VPCS>
```

Рисунок 73: Проверка доступности оконечных устройств

Раздел 4

4. Выводы

4.1 слайд 1

Я приобрел навыки по настройке службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.